

SOME MODEL-THEORETIC RESULTS ON THE 3-VALUED PARACONSISTENT FIRST-ORDER LOGIC **QCIORE**

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Abstract. The 3-valued paraconsistent logic **Ciore** was developed by Carnielli, Marcos and de Amo under the name **LFI2**, in the study of inconsistent databases from the point of view of *logics of formal inconsistency* (**LFI**s). They also considered a first-order version of **Ciore** called **LFI2***. The logic **Ciore** enjoys extreme features concerning propagation and retropropagation of the consistency operator: a formula is consistent if and only if some of its subformulas is consistent. In addition, **Ciore** is algebraizable in the sense of Blok and Pigozzi. On the other hand, the logic **LFI2*** satisfies a somewhat counter-intuitive property: the universal and the existential quantifier are inter-definable by means of the paraconsistent negation, as it happens in classical first-order logic with respect to the classical negation. This feature seems to be unnatural, given that both quantifiers have the classical meaning in **LFI2***, and that this logic does not satisfy the De Morgan laws with respect to its paraconsistent negation. The first goal of the present article is to introduce a first-order version of **Ciore** (which we call **QCiore**) preserving the spirit of **Ciore**, that is, without introducing unexpected relationships between the quantifiers. The second goal of the article is to adapt to **QCiore** the partial structures semantics for the first-order paraconsistent logic **LPT1** introduced by Coniglio and Silvestrini, which generalizes the semantic notion of *quasi-truth* considered by Mikeberg, da Costa and Chuaqui. Finally, some important results of classical Model Theory are obtained for this logic, such as Robinson's joint consistency theorem, amalgamation and interpolation. Although we focus on **QCiore**, this framework can be adapted to other 3-valued first-order **LFI**s.

§1. Introduction. A logic is paraconsistent if it has a negation which allows the existence of contradictory but nontrivial theories. The first systematic study of paraconsistent logics was given by da Costa, when presented in da Costa (1963) his well-known hierarchy C_n (for $n \geq 1$) of C -systems. His approach to paraconsistency is based on the idea that propositions are *prima facie dubious*, in the sense that a sentence φ and its negation $\neg\varphi$ can be assumed to hold simultaneously, without trivialization. The sentence $\varphi^\circ := \neg(\varphi \wedge \neg\varphi)$

Received: August 24, 2018.

2010 *Mathematics Subject Classification*: 03B53; 03C90.

Key words and phrases: paraconsistent logics, first-order logics, three-valued logics, logics of formal inconsistency, model theory.

indicates the *well-behavior* (in da Costa's terminology) of a sentence φ in the stronger system C_1 .¹ That sentence is used to express that a sentence φ is *reliable* (meaning that neither φ nor $\neg\varphi$ can be contradicted without trivializing). This being so, from φ , $\neg\varphi$ and φ° (as well as from $\neg\varphi$, $\neg\neg\varphi$ and φ°) every sentence ψ can be derived. In other words: a theory containing a contradiction $\{\varphi, \neg\varphi\}$ is not always deductively trivial (since φ represents, *prima facie*, dubious information, hence it could be contradictory) but any theory containing $\{\varphi, \neg\varphi, \varphi^\circ\}$ is always deductively trivial (since it contains a contradiction involving absolutely reliable information, which is an absurd).

The approach to paraconsistency of da Costa, nowadays known as the *Brazilian school of paraconsistency*, was naturally generalized by W. Carnielli and J. Marcos in Carnielli & Marcos (2002) with the notion of *Logics of Formal Inconsistency* (**LFIs**, for short). These logics are paraconsistent and, additionally, have a unary *consistency* connective $\circ$ (primitive or definable) that allows us to express the notion of consistency of sentences inside the object language. Thus, $\circ\varphi$ means that φ is *consistent*. The *Explosion Law* ('from φ and $\neg\varphi$ every ψ follows') does not hold in **LFIs**, as much as in any other paraconsistent logic. But some contradictions will cause deductive explosion: contradictions involving a *consistent* sentence lead to triviality – intuitively, and as in da Costa's *C*-systems, one can understand this situation as a contradiction involving well-established facts, or involving propositions that have conclusive favorable evidence (what da Costa called *well-behaved* sentences). In this sense, **LFIs** – including da Costa's systems – are logics that allow us to separate the sentences for which the Explosion Law hold, from those for which it does not hold. The only difference between da Costa's *C*-systems and **LFIs** in general is that, in the latter, the consistency operator is given by an unary connective $\circ$ which is possibly primitive while, in the former, it is defined in terms of the other connectives.

Positive classical logic (**CPL**⁺) has been assumed as a natural starting point from which many **LFIs** are defined, although some **LFIs** have been studied starting from other logics than **CPL**⁺. In this sense, **mbC** is the basic logic of formal inconsistency, which is an expansion of **CPL**⁺ by adding a negation and a consistency connective endowed with minimal properties in order to satisfy the definition of **LFIs**. In Carnielli & Marcos (2002), Carnielli, Coniglio, & Marcos (2007), and Carnielli & Coniglio (2016, chap. 2) some hierarchies of **LFIs** were presented, starting from **mbC** (in Carnielli & Marcos (2002) the starting point was the logic **bC**, the system obtained from **mbC** by adding (cf): $\neg\neg\varphi \rightarrow \varphi$). The system C_1 from da Costa is represented in these hierarchies by means of the logic **Cila**, an axiomatic extension of **mbC** in which $\circ\varphi$ is equivalent to φ° , that is: $\circ\varphi \leftrightarrow \neg(\varphi \wedge \neg\varphi)$ is a theorem in **Cila**.

An important feature of da Costa's systems C_n is that the well-behavior $(\cdot)^{(n)}$ is *propagated* throughout the connectives. This property is guaranteed by means of a specific axiom in each system C_n , namely:

$$(\varphi)^{(n)} \wedge \psi^{(n)} \rightarrow ((\varphi \wedge \psi)^{(n)} \wedge (\varphi \vee \psi)^{(n)} \wedge (\varphi \rightarrow \psi)^{(n)}).$$

In Béziau (1990) several alternatives to the logic C_1 were proposed, including a system called C_+ , obtained from C_1 by replacing the axiom of 'well-behavior propagation' by a stronger one, namely: $(\varphi^\circ \vee \psi^\circ) \rightarrow ((\varphi \wedge \psi)^\circ \wedge (\varphi \vee \psi)^\circ \wedge (\varphi \rightarrow \psi)^\circ)$. This system was studied in da Costa, Béziau, & Bueno (1995), under the name of C_1^+ , and it was called **Cilo** in the context of **LFIs**, presented as an axiomatic extension of **mbC**.

¹ For each $n \geq 1$ there is a sentence $\varphi^{(n)}$ denoting the well-behavior of φ . Note that $\varphi^{(1)}$ is φ° .

² In addition, the schema $\varphi^{(n)} \rightarrow (\neg\varphi)^{(n)}$ is proved in C_n from the other axioms.

such that $\circ\phi$ is equivalent to ϕ° . Observe that in C_1^+ it is enough that *one* (not *both*) of the immediate subformulas of $(\phi\#\psi)$ be well-behaved to guarantee the well-behavior of $(\phi\#\psi)$, for $\# \in \{\wedge, \vee, \rightarrow\}$. This property has, as a consequence, that the paraconsistent negation of C_1^+ satisfies more ‘classical’ properties than the paraconsistent negation of C_1 : for instance, $\neg(\phi \vee \neg\psi) \rightarrow (\neg\phi \wedge \psi)$ is a valid schema in C_1^+ but not in C_1 . Hence, C_1^+ is closer to classical logic than C_1 is, still being paraconsistent.³ Another justificative given in da Costa et al. (1995) for studying C_1^+ is that this logic would admit a nontrivial logic congruence (called *well-equivalence*) expressed by the validity of the formula $\psi(\alpha, \beta) := (\alpha^\circ \wedge \beta^\circ) \wedge (\alpha \leftrightarrow \beta)$. However, it was shown in (Carnielli & Coniglio, 2016, Prop. 4.3.7) that this formula does not determine a congruence, since reflexivity is not satisfied in general: $\psi(\alpha, \alpha)$ is a theorem of C_1^+ if and only if α° is a theorem of C_1^+ .

This problem can be overcome by requiring stronger properties to the well-behavior connective $(\cdot)^\circ$ (or to the consistency connective $\circ$, in the context of **LFIs**). Recall that **Cilo** is the version of C_1^+ in the language of **LFIs** (in the same manner that **Cila** corresponds to C_1). Let **Cio** be the system obtained from **Cilo** by eliminating the equivalence between α° and $\circ\alpha$ (thus ensuring that the primitive consistency operator is not definable in terms of the other ones).⁴ Let **Cior** be the extension of **Cio** by adding the following *consistency retropropagation* property: $(\circ(\phi \wedge \psi) \vee \circ(\phi \vee \psi) \vee \circ(\phi \rightarrow \psi)) \rightarrow (\circ\phi \vee \circ\psi)$ (see (Carnielli & Marcos, 2002, p. 67). Observe that, in **Cior**, $\circ(\phi\#\psi)$ is equivalent to $(\circ\phi \vee \circ\psi)$, for $\# \in \{\wedge, \vee, \rightarrow\}$.⁵ It can be proven that $\circ\phi \leftrightarrow \circ\neg\phi$ is a theorem of **Cior**. Finally, **Ciore** (see (Carnielli & Marcos, 2002, p. 68) is the extension of **Cior** by adding (ce): $\phi \rightarrow \neg\neg\phi$ (which produces the equivalence between ϕ and $\neg\neg\phi$). The logic **Ciore** was studied in Carnielli, Marcos, & de Amo (2000) under the name of **LFI2** (to be precise, they consider a primitive inconsistency connective $\bullet$ instead of a consistency connective $\circ$, but this approach is equivalent). As observed in Carnielli & Marcos (2002, Fact 3.82), contrary to C_1 and C_1^+ , the logic **Ciore** is algebraizable in the sense of Blok and Pigozzi (hence it *has*, indeed, a nontrivial logical congruence). More than this, because of the strong properties enjoyed by the consistency connective $\circ$ and by the paraconsistent negation, **Ciore** can be characterized by means of a 3-valued logical matrix, and it is maximal with respect to classical logic. The logic **Ciore** belongs to a family of 8,192 3-valued maximal paraconsistent logic considered in Marcos (2000) (see also Carnielli & Marcos (2002) and Carnielli et al. (2007)). The properties enjoyed by the logics in this family are remarkable, provided that in Carnielli et al. (2007) and Carnielli & Coniglio (2016) it was shown, in the style of the famous Dugundji’s theorem for modal logics, that a wide variety of **LFIs** extending **mbC** cannot be semantically characterized by a single finite logical matrix. Moreover, these systems are not algebraizable in any reasonable sense, including Blok and Pigozzi’s framework.⁶

³ It is worth noting that being ‘as close as possible’ to classical logic is one of the basic properties required for the C_n systems, see (da Costa, 1974, p. 498). Hence, this should be a fundamental feature for a ‘good’ paraconsistent logic, according to da Costa’s perspective.

⁴ Observe that $\circ\phi \rightarrow \neg(\phi \wedge \neg\phi)$ holds in **mbC** (and so in **bC**), hence the axiom to be eliminated from **Cilo** in order to get **Cio** is (cl): $\neg(\phi \wedge \neg\phi) \rightarrow \circ\phi$.

⁵ Notice that, by adding to **Cior** axiom (cl): $\neg(\phi \wedge \neg\phi) \rightarrow \circ\phi$, the resulting logic (called **Cilor**) is equivalent, in the language of **LFIs**, with the extension of C_1^+ obtained by adding retropropagation of well-behavior.

⁶ In (Carnielli & Coniglio, 2016, chap. 6) these **LFIs** were characterized in terms of nondeterministic semantics called *Fidel structures* and *swap structures*.

The logic **Ciore** enjoys extreme features concerning propagation and retropropagation of the consistency operator: the consistency of a formula φ is equivalent in **Ciore** to the consistency of some of its proper subformulas. In particular, if p is a propositional variable that occurs in φ then $\circ p \rightarrow \circ \varphi$ is a theorem in **Ciore**. Equivalently, a formula is inconsistent (that is, it gets the ‘inconsistent’ truth-value $\frac{1}{2}$ under a given valuation) if and only if every subformula (and so every propositional variable occurring in the formula) gets the truth-value $\frac{1}{2}$ under such valuation. These properties, together with the motivations behind its definition (related to C_1^+ , as discussed above), give to this logic a privileged status in the family of 8K 3-valued maximal **LFI**s mentioned above.

Before this family, other 3-valued propositional paraconsistent logics were proposed in the literature. The *Logic of Paradox* **LP** introduced in Priest (1979) is probably the most renowned of them. Another well-known 3-valued paraconsistent logic is the logic **J3**, introduced in D’Ottaviano & da Costa (1970). While **LP** is not an **LFI** (since it is not finitely trivalizable), it can be easily proven that the logic **J3** is an **LFI** (see Carnielli & Marcos (2002)). In Carnielli et al. (2000) it was proposed the use of **J3**, now seen as an **LFI** rebaptized as **LFI1**, as a suitable tool for modeling processes in databases with contradictory information. Besides this, they defined a first-order version of **LFI1** called **LFI1***, which happens to be equivalent to the first-order version of **J3** studied in the pioneering articles D’Ottaviano (1982, 1985a,b, 1987).

The logic **LFI1*** is also equivalent to the first-order logic **LPT1** introduced in Coniglio & Silvestrini (2014). Indeed, **LPT1** is based on the 3-valued propositional logic **MPT**, which was proved to be functionally equivalent to **J3** (and therefore also to **LFI1**). The aim of Coniglio & Silvestrini (2014) was to establish a 3-valued logical matrix associated to *partial structures* (also called *pragmatic structures*) which constitute a generalization of the semantic notion of *quasi-truth*, as introduced in Mikenberg, da Costa, & Chuaqui (1986). Partial structures are first-order structures where each n -ary predicate R is interpreted as a triple $\langle R_+, R_-, R_u \rangle$ of pairwise disjoint sets of n -uples over the domain of the structure such that R_+ is the set of tuples which actually belong to R , R_- is the set of tuples which actually do not belong to R , and R_u is the set of tuples whose status is dubious or contradictory. The approach to quasi-truth by means of triples presented in Coniglio & Silvestrini (2014) is useful since, despite being clear from the intuitive point of view, it is equivalent (at the propositional level) to the twist structures semantics for **LFI1** given in Coniglio, Figallo-Orellano, & Golzio (2018) (this can be easily proven by adapting the arguments for **Ciore** to be given in §5 below). This is an important feature, since twist-structures characterize the algebraic structures behind an algebraizable logic.

Besides **LFI1***, a first-order version of **LFI2** (that is, of **Ciore**) called **LFI2*** was studied in Carnielli et al. (2000). However, the logic **LFI2*** satisfies a counter-intuitive property: the universal and the existential quantifier are inter-definable by means of the paraconsistent negation as it happens in **LFI1*** (and in classical first-order logic with respect to the classical negation). While this property is expected in **LFI1*** (given that **LFI1** satisfies the De Morgan laws with respect to its paraconsistent negation), the inter-definability of quantifiers by means of the paraconsistent negation seems to be unnatural in any first-order version of **Ciore** in which the quantifiers have the classical meaning, given that this logic does not satisfy the De Morgan laws with respect to its paraconsistent negation. Some undesirable consequences of the inter-definability requirement in **LFI2*** will be analyzed in §7.1.

The first goal of the present article is to introduce a first-order version of **Ciore** (which we call **QCiore**) preserving the spirit of **Ciore**, that is, without introducing unexpected

relationships between the quantifiers. Indeed, in §7.1 it will be shown that the inter-definability of quantifiers by means of the paraconsistent negation does not hold in **QCiore**. The second goal of the article is to adapt the very intuitive partial structures semantics of **LPT1** to **QCiore**. As it happens with **LPT1**, this triples semantics is equivalent (at the propositional level) to the twist structures semantics for **Ciore**, which characterizes the algebraic counterpart of **Ciore** (see §5). Finally, based on this nice semantical framework, and after proving adequacy of **QCiore** w.r.t. pragmatic structures, some important results of classical Model Theory are obtained for this logic, such as Robinson's joint consistency theorem, amalgamation and interpolation. Although we focus on **QCiore**, this framework can be adapted to other 3-valued first-order **LFIs**.

§2. Preliminaries. In this section, a Hilbert-style presentation for **Ciore** will be given. Let $\mathcal{L}$ be the propositional language defined over the propositional signature $\Sigma = \{\wedge, \vee, \rightarrow, \neg, \circ\}$ where the formulas are constructed as usual from a denumerable set $\{p_1, p_2, \dots\}$ of propositional variables.

The propositional logic **Ciore** is defined over the language $\mathcal{L}$ by the following Hilbert system (as usual, $\alpha \leftrightarrow \beta$ denotes the formula $(\alpha \rightarrow \beta) \wedge (\beta \rightarrow \alpha)$):

Axiom schemata:

$$\begin{array}{ll}
 \alpha \rightarrow (\beta \rightarrow \alpha) & (\text{Ax1}) \\
 (\alpha \rightarrow (\beta \rightarrow \gamma)) \rightarrow ((\alpha \rightarrow \beta) \rightarrow (\alpha \rightarrow \gamma)) & (\text{Ax2}) \\
 \alpha \rightarrow (\beta \rightarrow (\alpha \wedge \beta)) & (\text{Ax3}) \\
 (\alpha \wedge \beta) \rightarrow \alpha & (\text{Ax4}) \\
 (\alpha \wedge \beta) \rightarrow \beta & (\text{Ax5}) \\
 \alpha \rightarrow (\alpha \vee \beta) & (\text{Ax6}) \\
 \beta \rightarrow (\alpha \vee \beta) & (\text{Ax7}) \\
 (\alpha \rightarrow \gamma) \rightarrow ((\beta \rightarrow \gamma) \rightarrow ((\alpha \vee \beta) \rightarrow \gamma)) & (\text{Ax8}) \\
 (\alpha \rightarrow \beta) \vee \alpha & (\text{Ax9}) \\
 \alpha \vee \neg \alpha & (\text{Ax10}) \\
 \alpha \rightarrow (\alpha \rightarrow (\neg \alpha \rightarrow \beta)) & (\text{bc1}) \\
 \neg \alpha \rightarrow (\alpha \wedge \neg \alpha) & (\text{ci}) \\
 \neg \neg \alpha \leftrightarrow \alpha & (\text{cef}) \\
 (\circ \alpha \vee \circ \beta) \leftrightarrow \circ(\alpha \wedge \beta) & (\text{cor}_1) \\
 (\circ \alpha \vee \circ \beta) \leftrightarrow \circ(\alpha \vee \beta) & (\text{cor}_2) \\
 (\circ \alpha \vee \circ \beta) \leftrightarrow \circ(\alpha \rightarrow \beta) & (\text{cor}_3)
 \end{array}$$

Inference rule:

$$(\text{MP}) \quad \frac{\alpha \quad \alpha \rightarrow \beta}{\beta}$$

The (Tarskian) consequence relation obtained from the Hilbert calculus for **Ciore** will be denoted by $\vdash_{\text{Ciore}}$. Observe that axioms (Ax1)-(Ax9) plus (MP) constitute a Hilbert calculus for positive classical logic (**CPL**⁺). As observed in the Introduction, the logic

Ciore was proposed as a suitable extension of **Cio** which is algebraizable in the sense of Blok and Pigozzi. Moreover, the properties of **Ciore** are so strong that this logic can be characterized by a 3-valued logical matrix.

THEOREM 2.1. (See, for instance, (Carnielli & Coniglio, 2016, Theorem 4.4.29)) *The system **Ciore** is sound and complete with respect to the following three valued matrix $M_e = \langle \langle M, \wedge, \vee, \rightarrow, \neg, \circ \rangle, D \rangle$ over the signature Σ with domain $M = \{1, \frac{1}{2}, 0\}$ and set of designated values $D = \{1, \frac{1}{2}\}$ such that the truth-tables associated to each connective are displayed in Figure 1.*

$\wedge$	1	$\frac{1}{2}$	0
1	1	1	0
$\frac{1}{2}$	1	$\frac{1}{2}$	0
0	0	0	0

$\vee$	1	$\frac{1}{2}$	0
1	1	1	1
$\frac{1}{2}$	1	$\frac{1}{2}$	1
0	1	1	0

$\rightarrow$	1	$\frac{1}{2}$	0
1	1	1	0
$\frac{1}{2}$	1	$\frac{1}{2}$	0
0	1	1	1

$\neg$	$\circ$
1	0
$\frac{1}{2}$	$\frac{1}{2}$
0	1

Fig. 1. Truth-tables for **Ciore**.

As mentioned in the Introduction, the logic associated with the logical matrix M_e was studied in Carnielli et al. (2000) under the name of **LFI2** in the context of inconsistent databases,⁷ being called **Ciore** in Carnielli & Marcos (2002). In the following result, we exhibit a list of theorems of **Ciore** that we shall use along this article. We will write $\sim\alpha$ as an abbreviation of $\neg\alpha \wedge \circ\alpha$ to denote the strong (or classical) negation definable in **Ciore**. Note that $\sim 0 = 1$ and $\sim 1 = \frac{1}{2} = 0$ in M_e .

THEOREM 2.2. *The following schemas are theorems of **Ciore**.*

- (i) $\alpha \rightarrow \alpha$,
- (ii) $(\alpha \rightarrow \beta) \rightarrow (\sim\beta \rightarrow \sim\alpha)$,
- (iii) $(\alpha \wedge \neg\alpha) \leftrightarrow \sim\circ\alpha$,
- (iv) $(\alpha \wedge \neg\alpha) \leftrightarrow \neg\circ\alpha$,
- (v) $\circ\circ\alpha$,
- (vi) $\circ\alpha \leftrightarrow \circ\neg\alpha$,
- (vii) $(\sim\alpha \vee \sim\beta) \leftrightarrow \sim(\alpha \wedge \beta)$, (viii) $((\alpha \wedge \neg\alpha) \wedge (\beta \wedge \neg\beta)) \leftrightarrow ((\alpha \wedge \beta) \wedge \neg(\alpha \wedge \beta))$,
- (ix) $((\alpha \wedge (\beta \wedge \circ\beta)) \vee ((\alpha \wedge \circ\alpha) \wedge \beta)) \leftrightarrow ((\alpha \wedge \beta) \wedge \circ(\alpha \wedge \beta))$,
- (x) $((\alpha \wedge \circ\alpha) \vee (\beta \wedge \circ\beta) \vee ((\alpha \wedge \neg\alpha) \wedge \sim\beta) \vee (\sim\alpha \wedge (\beta \wedge \neg\beta))) \leftrightarrow ((\alpha \vee \beta) \wedge \circ(\alpha \vee \beta))$,
- (xi) $(\sim\alpha \wedge \sim\beta) \leftrightarrow \sim(\alpha \vee \beta)$,
- (xii) $((\alpha \wedge \neg\alpha) \wedge (\beta \wedge \neg\beta)) \leftrightarrow ((\alpha \vee \beta) \wedge \neg(\alpha \vee \beta))$,
- (xiii) $(\sim\alpha \vee (\beta \wedge \circ\beta)) \leftrightarrow ((\alpha \rightarrow \beta) \wedge \circ(\alpha \rightarrow \beta))$,
- (xiv) $(\alpha \wedge \sim\beta) \leftrightarrow \sim(\alpha \rightarrow \beta)$,
- (xv) $((\alpha \wedge \neg\alpha) \wedge (\beta \wedge \neg\beta)) \leftrightarrow ((\alpha \rightarrow \beta) \wedge \neg(\alpha \rightarrow \beta))$.

Proof. It is routine to check that all these formulas are tautologies in **Ciore** by observing that, for every $x \in M$: $x = 1$ iff $x \wedge \circ x \in D$; $x = 0$ iff $\neg x \wedge \circ x \in D$; and $x = \frac{1}{2}$ iff $x \wedge \neg x \in D$. The result follows by Theorem 2.1. $\square$

§3. Partial relations. In order to define the semantical framework for **QCiore**, the first-order version of **Ciore** to be introduced in §4, it will be convenient to consider partial relations, defined in terms of triples. Note that we could have used the more standard

⁷ The authors use an inconsistency connective instead of a consistency connective, replacing so, in M_e , $\circ$ by the inconsistency operator $\bullet$ ($\bullet(\frac{1}{2}) = 1$ and $\bullet(0) = \bullet(1) = 0$). Note that $\circ(x) = \neg\bullet(x)$ and $\bullet(x) = \neg\circ(x)$.

structure of pairs of extension/anti-extension used, for example, by Priest in the **LP**-models for the first-order version of the paraconsistent logic **LP** (see, for instance, Priest (1979, 2006)). The semantical approach ‘by pairs’ coincides with the so-called *twist structures*, introduced by Fidel and Vakarelov and anticipated by Kalman (a more detailed discussion on this topic will be presented in §5). The triples approach was adopted here because it is closely related to the quasi-truth theory, as explained in the Introduction. This representation is quite clear from the conceptual point of view, keeping all the semantic information about the sentences (the truth, the false and the dubious) on separated components. In addition, pragmatic structures provide a formal framework where definitions and proofs are more clean and much more simple to deal with. Some proofs (Henkin Lemma, for instance) and definitions (elementary substructure) are much more complex to write by using pairs instead of triples. Moreover, partial relations can be defined in a very natural way by using a simple and intuitive algebraic construction of triples, to be defined now.

Let $\mathbf{3} = \{0, \frac{1}{2}, 1\}$ and consider the algebraic structure $\mathbf{3} = \langle \mathbf{3}; \vee, \wedge, \rightarrow, \neg, \circ \rangle$ underlying the 3-valued logical matrix of **Ciore**, where the operations are defined as in Theorem 2.1. Let X be a nonempty set. A *triple over X* is a map $r : X \rightarrow \mathbf{3}$. If r is a triple over X we write $r = \langle r_{\oplus}, r_{\ominus}, r_{\circ} \rangle$ where $r_{\oplus} = r^{-1}(1)$, $r_{\circ} = r^{-1}(\frac{1}{2})$, $r_{\ominus} = r^{-1}(0)$. As usual, we denote by $\mathbf{3}^X$ the set of all triples over X .

REMARK 3.1. *Clearly, the set $\mathbf{3}^X$ of triples inherits the algebraic structure of $\mathbf{3}$, where the operations are defined pointwise. Namely, $(r \div u)(x) := r(x) \div u(x)$ and $(*r)(x) := *(r(x))$ for $\div \in \{\wedge, \vee, \rightarrow\}$, $* \in \{\neg, \circ\}$, $x \in X$ and $r, u \in \mathbf{3}^X$.*

*In a sense, $\mathbf{3}^X$ generalizes the powerset $\mathbf{2}^X$ (seen as a Boolean algebra), and so $\mathbf{3}^X$ is a kind of 3-valued powerset, endowed with the 3-valued algebraic structure over Σ inherited from the 3-valued matrix of **Ciore**. Observe that this construction can be adapted to any 3-valued logical matrix, see Remark 3.3. The generalization of this technique to n -valued matrices is an interesting issue, that will be analyzed in a future article.*

It is useful to see that the operations in the Σ -algebra $\mathbf{3}^X$ of triples can be expressed in terms of the Boolean operators in $\mathbf{2}^X$, as the following result shows.

PROPOSITION 3.2. *Let $r = (r_{\oplus}, r_{\ominus}, r_{\circ})$ and $u = (u_{\oplus}, u_{\ominus}, u_{\circ})$ be two triples over X . Then:*

- (i) $r \wedge u = ((r_{\oplus} \cap u_{\oplus}) \cup (r_{\oplus} \cap u_{\circ}) \cup (r_{\circ} \cap u_{\oplus}), r_{\ominus} \cup u_{\ominus}, r_{\circ} \cap u_{\circ})$,
- (ii) $r \vee u = (r_{\oplus} \cup u_{\oplus} \cup (r_{\circ} \cap u_{\ominus}) \cup (r_{\ominus} \cup u_{\circ}), r_{\ominus} \cap u_{\ominus}, r_{\circ} \cap u_{\circ})$,
- (iii) $r \rightarrow u = (r_{\ominus} \cup u_{\oplus} \cup (r_{\oplus} \cap u_{\circ}), (r_{\oplus} \cup r_{\circ}) \cap u_{\ominus}, r_{\circ} \cap u_{\circ})$,
- (iv) $\neg r = (r_{\ominus}, r_{\oplus}, r_{\circ})$,
- (v) $r \circ u = (r_{\oplus} \cup r_{\ominus}, r_{\circ}, \emptyset)$.

Proof. Here, only item (iii) will be proved. First, $x \in (r \rightarrow u)_{\oplus}$ iff $r(x) \rightarrow u(x) = 1$ iff $r(x) = 0$ or $u(x) = 1$ iff $x \in r_{\ominus} \cup r_{\oplus}$. Secondly, $x \in (r \rightarrow u)_{\ominus}$ iff $r(x) \rightarrow u(x) = 0$ iff $r(x) \in \{1, \frac{1}{2}\}$ and $u(x) = 0$ iff $x \in (r_{\oplus} \cup r_{\circ}) \cap u_{\ominus}$. Finally, $x \in (r \rightarrow u)_{\circ}$ iff $r(x) \rightarrow u(x) = \frac{1}{2}$ iff $r(x) \in \{1, \frac{1}{2}\}$ and $u(x) = \frac{1}{2}$ iff $x \in (r_{\oplus} \cup r_{\circ}) \cap u_{\oplus}$. $\square$

The notion of *Partial Relation* introduced in Mikenberg et al. (1986) can be represented by the notion of triple. We interpret the nature of a triple $r = (r_{\oplus}, r_{\ominus}, r_{\circ})$ over $X \neq \emptyset$ in the context of **QCiore** as being a kind of 3-valued fuzzy set such that:

- $r_{\oplus}$ is the set of individuals which belongs to r ,
- $r_{\ominus}$ is the set of individuals which do not belong to r ,
- $r_{\circ}$ is the set of individuals whose status w.r.t. r is dubious or contradictory.

It is not difficult to see that, if $r \in \mathbf{3}^X$, then: (1) $r_{\oplus} \cup r_{\ominus} \cup r_{\circ} = X$; and (2) $r_{\oplus} \cap r_{\ominus} = r_{\oplus} \cap r_{\circ} = r_{\ominus} \cap r_{\circ} = \emptyset$. Conversely, if r_1, r_2 and r_3 are subsets of $X \neq \emptyset$ such that (1) $r_1 \cup r_2 \cup r_3 = X$ and (2) $r_1 \cap r_2 = r_1 \cap r_3 = r_2 \cap r_3 = \emptyset$, then there exists a unique $r \in \mathbf{3}^X$ such that $r = (r_1, r_2, r_3)$.

REMARK 3.3. *It is worth noting that, by considering other 3-valued logical matrices (possibly over different signatures), the corresponding operations will be induced in $\mathbf{3}^X$ accordingly. For example:*

(1) Let $\mathcal{M}_{\mathbf{P1}} = \langle \langle \mathbf{3}, \rightarrow, \neg \rangle, \{1, \frac{1}{2}\} \rangle$ be the 3-valued logical matrix associated to Sette's logic **P1** (see Sette (1973)), where the truth-tables are displayed in Figure 2.

$\rightarrow$	1	$\frac{1}{2}$	0
1	1	1	0
$\frac{1}{2}$	1	1	0
0	1	1	1

	$\neg$
1	0
$\frac{1}{2}$	1
0	1

Fig. 2. Truth-tables for **P1**.

Then, for $r, u \in \mathbf{3}^X$, we obtain the following:

$$\neg r = (r_{\ominus} \cup r_{\circ}, r_{\oplus}, \emptyset)$$

$$r \rightarrow u = (r_{\ominus} \cup u_{\oplus} \cup u_{\circ}, (r_{\oplus} \cup r_{\circ}) \cap u_{\ominus}, \emptyset)$$

This coincides with the operations over triples for **P1** proposed in (Carnielli & Coniglio, 2016, Definition 7.10.9).

(2) In (Carnielli & Coniglio, 2016, chap. 4) a new presentation of **LFII** (or **J3**) called **LFII'** was introduced, by taking as primitive the connective $\circ$ (the consistency operator) instead of $\bullet$ (the inconsistency operator), together with a suitable Hilbert calculus called **LFII_o**. Let $\mathcal{M}_{\mathbf{LFII}'} = \langle \langle \mathbf{3}, \wedge, \vee, \rightarrow, \neg, \circ \rangle, \{1, \frac{1}{2}\} \rangle$ be the 3-valued logical matrix associated to **LFII_o**, where the truth-tables are displayed in Figure 3.

$\wedge$	1	$\frac{1}{2}$	0
1	1	$\frac{1}{2}$	0
$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	0
0	0	0	0

$\vee$	1	$\frac{1}{2}$	0
1	1	1	1
$\frac{1}{2}$	1	$\frac{1}{2}$	$\frac{1}{2}$
0	1	$\frac{1}{2}$	0

$\rightarrow$	1	$\frac{1}{2}$	0
1	1	$\frac{1}{2}$	0
$\frac{1}{2}$	1	$\frac{1}{2}$	0
0	1	1	1

	$\neg$	$\circ$
1	0	1
$\frac{1}{2}$	$\frac{1}{2}$	0
0	1	1

Fig. 3. Truth-tables for **LFII'**.

As in the case of **P1**, it can be seen that the operations over $\mathbf{3}^X$ induced by the truth-tables above coincide with the operations over triples for **LFII'**/**LFII_o** proposed in (Carnielli & Coniglio, 2016, Definition 7.9.2) which, by its turn, were adapted from the operations over triples for **MPT** given in Coniglio & Silvestrini (2014). As observed in Carnielli & Coniglio (2016) and Coniglio & Silvestrini (2014), the logics **LFII'** and **MPT** coincide, up to language, with **LFII** and **J3**.

§4. The paraconsistent first-order logic **QCiore.** In this section the logic **QCiore** will be introduced, as a natural first-order version of **Ciore**, with semantics based on the notion of triples (or partial relations) analyzed in the previous section. The axiomatization of **QCiore** is based on the approach to first-order **LFI**s given in Carnielli, Coniglio, Podiacki, & Rodrigues (2014) and (Carnielli & Coniglio, 2016, chap. 7). Hence, **QCiore** will be defined as an axiomatic extension of **QmbC**, the first-order version of **mbC** proposed in these references (the axiom: “ $\varphi \rightarrow \psi$, if φ is a variant of ψ ” of **QmbC** can be derived from the other axioms of **QCiore**).

Recall that a first-order signature $\Theta = \langle \mathcal{P}, \mathcal{F}, \mathcal{C} \rangle$ is composed of the following elements: a set $\mathcal{P} = \bigcup_{n \in \mathbb{N}} \mathcal{P}_n$ such that, for each $n \geq 1$, $\mathcal{P}_n$ is a set of predicate symbols of arity n ; a set $\mathcal{F} = \bigcup_{n \in \mathbb{N}} \mathcal{F}_n$ such that, for each $n \geq 1$, $\mathcal{F}_n$ is a set of function symbols of arity n ; and a set $\mathcal{C}$ of individual constants. Let $\mathcal{L}(\Theta)$ be the first-order language defined as usual from the connectives $\wedge, \vee, \rightarrow, \neg, \circ$, the quantifiers $\forall, \exists$, a denumerable set $\mathcal{V}$ of individual variables and a given first-order signature Θ . We denote by $Sent(\Theta)$ the set of sentences (formulas without free-variables) over the signature Θ . If φ is a formula, x is a variable and t is a term, then $\varphi(t/x)$ will denote the formula that results from φ by replacing simultaneously all the free occurrences of the variable x by t .

DEFINITION 4.1. *Let Θ be a first-order signature. The logic **QCiore** on the language $\mathcal{L}(\Theta)$ is defined as the Hilbert calculus obtained by extending **Ciore** (expressed in the language $\mathcal{L}(\Theta)$) by adding the following:*

Axiom schemata:

$$\varphi(t/x) \rightarrow \exists x\varphi, \text{ if } t \text{ is a term free for } x \text{ in } \varphi \quad (\text{Ax11})$$

$$\forall x\varphi \rightarrow \varphi(t/x), \text{ if } t \text{ is a term free for } x \text{ in } \varphi \quad (\text{Ax12})$$

$$\circ\exists x\varphi \leftrightarrow \exists x\circ\varphi, \quad (\text{Ax13})$$

$$\circ\forall x\varphi \leftrightarrow \forall x\circ\varphi, \quad (\text{Ax14})$$

Inference rules

$$(\forall\text{-In}) \frac{\alpha \rightarrow \beta}{\alpha \rightarrow \forall x\beta} \quad \text{if } x \text{ is not free in } \alpha$$

$$(\exists\text{-In}) \frac{\alpha \rightarrow \beta}{\exists x\alpha \rightarrow \beta} \quad \text{if } x \text{ is not free in } \beta$$

The (Tarskian) consequence relation obtained from the Hilbert calculus for **QCiore** will be denoted by $\vdash_{\text{QCiore}}$.

THEOREM 4.2. ***QCiore** proves the following schemas:*

- (i) $\sim\exists x\alpha \leftrightarrow \forall x\sim\alpha$,
- (ii) $\sim\forall x\alpha \leftrightarrow \exists x\sim\alpha$,
- (iii) $(\exists x\varphi \wedge \neg\exists x\varphi) \rightarrow \forall x(\varphi \wedge \neg\varphi)$,
- (iv) $(\forall x\varphi \wedge \neg\forall x\varphi) \rightarrow \forall x(\varphi \wedge \neg\varphi)$.

Proof. Items (i)–(ii) can be proved from the similar results obtained for **QmbC** in (Carnielli & Coniglio, 2016, chap. 7).

(iii):

1. $\exists x \circ \varphi \rightarrow \circ \exists x \varphi$	(Ax13)
2. $\sim \circ \exists x \varphi \rightarrow \sim \exists x \circ \varphi$	1, Theorem 2.2(ii)
3. $(\exists x \varphi \wedge \neg \exists x \varphi) \rightarrow \sim \circ \exists x \varphi$	Theorem 2.2(iii)
4. $(\exists x \varphi \wedge \neg \exists x \varphi) \rightarrow \sim \exists x \circ \varphi$	3, 2, transitivity of $\rightarrow$
5. $\sim \exists x \circ \varphi \rightarrow \forall x \sim \circ \varphi$	Item (ii)
6. $(\exists x \varphi \wedge \neg \exists x \varphi) \rightarrow \forall x \sim \circ \varphi$	4, 5, transitivity of $\rightarrow$
7. $\forall x \sim \circ \varphi \rightarrow \sim \circ \varphi$	(Ax12)
8. $\sim \circ \varphi \rightarrow (\varphi \wedge \neg \varphi)$	Theorem 2.2(iii)
9. $\forall x \sim \circ \varphi \rightarrow (\varphi \wedge \neg \varphi)$	7, 8, transitivity of $\rightarrow$
10. $\forall x \sim \circ \varphi \rightarrow \forall x (\varphi \wedge \neg \varphi)$	9, ($\forall$ -In)
11. $(\exists x \varphi \wedge \neg \exists x \varphi) \rightarrow \forall x (\varphi \wedge \neg \varphi)$	6, 10, transitivity of $\rightarrow$

(iv):

1. $\exists x \circ \varphi \rightarrow \circ \forall x \varphi$	(Ax14)
2. $\sim \circ \forall x \varphi \rightarrow \sim \exists x \circ \varphi$	1, Theorem 2.2(ii)
3. $(\forall x \varphi \wedge \neg \forall x \varphi) \rightarrow \sim \circ \forall x \varphi$	Theorem 2.2(iii)
4. $(\forall x \varphi \wedge \neg \forall x \varphi) \rightarrow \sim \exists x \circ \varphi$	3, 2, transitivity of $\rightarrow$
5. $\sim \exists x \circ \varphi \rightarrow \forall x \sim \circ \varphi$	Item (ii)
6. $(\forall x \varphi \wedge \neg \forall x \varphi) \rightarrow \forall x \sim \circ \varphi$	4, 5, transitivity of $\rightarrow$
7. $\forall x \sim \circ \varphi \rightarrow \sim \circ \varphi$	(Ax12)
8. $\sim \circ \varphi \rightarrow (\varphi \wedge \neg \varphi)$	Theorem 2.2(iii)
9. $\forall x \sim \circ \varphi \rightarrow (\varphi \wedge \neg \varphi)$	7, 8, transitivity of $\rightarrow$
10. $\forall x \sim \circ \varphi \rightarrow \forall x (\varphi \wedge \neg \varphi)$	9, ($\forall$ -In)
11. $(\forall x \varphi \wedge \neg \forall x \varphi) \rightarrow \forall x (\varphi \wedge \neg \varphi)$	6, 10, transitivity of $\rightarrow$

□

THEOREM 4.3. *The Generalization rule is derivable in QCiore, i.e.,*

	(Gen) $\alpha \vdash_{\text{QCiore}} \forall x \alpha$	
Proof.	1. α	Hyp.
	2. $\alpha \rightarrow ((\forall x \alpha \rightarrow \forall x \alpha) \rightarrow \alpha)$	(Ax 1)
	3. $(\forall x \alpha \rightarrow \forall x \alpha) \rightarrow \alpha$	1, 2, (MP)
	4. $(\forall x \alpha \rightarrow \forall x \alpha) \rightarrow \forall x \alpha$	3, ($\forall$ -In)
	5. $\forall x \alpha \rightarrow \forall x \alpha$	Theorem 2.2(i)
	6. $\forall x \alpha$	4, 5, (MP)

□

Finally, it is worth mentioning that it is possible to obtain in QCiore a weak version of the *deduction metatheorem*, as it happens with first-order classical logic, whose demonstration is identical to the one given for QmbC in (Carnielli & Coniglio, 2016, chap. 7).

THEOREM 4.4. (*Weak deduction metatheorem for QCiore*) *Suppose that there exists in QCiore a derivation of ψ from $\Gamma \cup \{\varphi\}$ such that no application of the rules ($\exists$ -In) and ($\forall$ -In) have, as their quantified variables, free variables of φ (in particular, this holds when φ is a sentence). Then $\Gamma \vdash_{\text{QCiore}} \varphi \rightarrow \psi$.*

§5. Triples and twist structures for Ciore. The so-called *twist structures* constitute an algebraic semantics for logics in terms of pairs of elements of a given class of algebras. The intuition behind twist structures semantics is that a pair (a, b) represents a truth-value for a formula φ , while b represents a truth-value for the negation of φ . Twist structures semantics were independently introduced in Fidel (1978) and in Vakarelov

(1977), in order to obtain a representation of Nelson algebras in terms of twist structures over Heyting algebras. As observed in Cignoli (1986), Fidel-Vakarelov structures are a particular case of a general construction proposed by Kalman in 1958 (see Kalman (1958)). In (Coniglio et al., 2018, Definition 9.18) the following twist structures semantics for **Ciore** was presented:

DEFINITION 5.1. *Let $\mathcal{A} = \langle A, \sim, \supset, \sqcup, \sqcap, 0, 1 \rangle$ be a Boolean algebra and let $\mathbb{P}_{\mathcal{A}} = \{(a, b) \in A \times A : a \sqcup b = 1\}$. A twist structure for **Ciore** over $\mathcal{A}$ is an algebra $\mathcal{P}_{\mathcal{A}}$ over the signature $\{\wedge, \vee, \rightarrow, \neg, \circ\}$ with domain $\mathbb{P}_{\mathcal{A}}$ such that the operations are defined as follows, for every $(z_1, z_2), (w_1, w_2) \in \mathbb{P}_{\mathcal{A}}$:*

- (i) $(z_1, z_2) \wedge (w_1, w_2) = (z_1 \sqcap w_1, (z_1 \sqcap w_1) \supset ((z_1 \sqcap z_2) \sqcap (w_1 \sqcap w_2)))$;
- (ii) $(z_1, z_2) \vee (w_1, w_2) = (z_1 \sqcup w_1, (z_1 \sqcup w_1) \supset ((z_1 \sqcap z_2) \sqcap (w_1 \sqcap w_2)))$;
- (iii) $(z_1, z_2) \rightarrow (w_1, w_2) = (z_1 \supset w_1, (z_1 \supset w_1) \supset ((z_1 \sqcap z_2) \sqcap (w_1 \sqcap w_2)))$;
- (iv) $\neg(z_1, z_2) = (z_2, z_1)$;
- (v) $\circ(z_1, z_2) = (\sim(z_1 \sqcap z_2), z_1 \sqcap z_2)$.

In what follows, it will be proven that the twist structures for **Ciore** are equivalent to the algebras of triples presented in §3. In order to do this, it will be observed that each algebra $\mathbf{3}^X$, which is formed by triples over the Boolean algebra $\wp(X)$ of all the subsets of X (by Proposition 3.2), can be generalized to any Boolean algebra.

DEFINITION 5.2. *Let $\mathcal{A}$ be a Boolean algebra and let $\mathbb{T}_{\mathcal{A}} = \{(a, b, c) \in A \times A \times A : a \sqcap b = a \sqcap c = b \sqcap c = 0, \text{ and } a \sqcup b \sqcup c = 1\}$. A structure of triples for **Ciore** over $\mathcal{A}$ is an algebra $\mathcal{T}_{\mathcal{A}}$ over the signature $\{\wedge, \vee, \rightarrow, \neg, \circ\}$ with domain $\mathbb{T}_{\mathcal{A}}$ such that the operations are defined as in Proposition 3.2, by changing $\cap$ and $\cup$ by $\sqcap$ and $\sqcup$, respectively.*

Thus, if $z = (z_1, z_2, z_3)$ and $w = (w_1, w_2, w_3)$ are in $\mathbb{T}_{\mathcal{A}}$ then, for instance, $\neg z = (z_2, z_1, z_3)$ and $z \rightarrow w = (z_2 \sqcup w_1, (z_1 \sqcup z_3) \sqcap w_2, (z_1 \sqcup z_3) \sqcap w_3)$. It is clear that, given a nonempty set X , the algebra $\mathbb{T}_{\wp(X)}$ is isomorphic to $\mathbf{3}^X$, by Proposition 3.2. Hence, Definition 5.2 generalizes the algebras of triples $\mathbf{3}^X$ from $\wp(X)$ to any Boolean algebra $\mathcal{A}$.

REMARK 5.3. *Let $\mathcal{A}$ be a Boolean algebra. For every $z \in \mathbb{P}_{\mathcal{A}}$ and every $w \in \mathbb{T}_{\mathcal{A}}$ let $\perp_{\mathcal{A}}^P = (z \wedge \neg z) \wedge \circ z$ and $\perp_{\mathcal{A}}^T = (w \wedge \neg w) \wedge \circ w$. By Definitions 5.1 and 5.2 it is easy to see that $\perp_{\mathcal{A}}^P = (0, 1)$ and $\perp_{\mathcal{A}}^T = (0, 1, 0)$, hence the choice of z and w for defining $\perp_{\mathcal{A}}^P$ and $\perp_{\mathcal{A}}^T$ is irrelevant. Moreover, $\circ z = (z \wedge \neg z) \rightarrow \perp_{\mathcal{A}}^P$ and $\circ w = (w \wedge \neg w) \rightarrow \perp_{\mathcal{A}}^T$ for every $z \in \mathbb{P}_{\mathcal{A}}$ and every $w \in \mathbb{T}_{\mathcal{A}}$.*

PROPOSITION 5.4. *Let $\mathcal{A}$ be a Boolean algebra. Then, the function $\dagger : \mathbb{T}_{\mathcal{A}} \rightarrow \mathbb{P}_{\mathcal{A}}$ given by $\dagger(z_1, z_2, z_3) = (z_1 \sqcup z_3, z_2 \sqcup z_3)$ is an isomorphism between $\mathcal{T}_{\mathcal{A}}$ and $\mathcal{P}_{\mathcal{A}}$. The inverse of $\dagger$ is given by $\ddagger : \mathbb{P}_{\mathcal{A}} \rightarrow \mathbb{T}_{\mathcal{A}}$ such that $\ddagger(z_1, z_2) = (z_1 \sqcap \sim z_2, z_2 \sqcap \sim z_1, z_1 \sqcap z_2)$.*

Proof. Straightforward. Observe that, by Remark 5.3, the fact that $\dagger(\circ z) = \circ \dagger(z)$ follows from the fact that $\dagger(\neg z) = \neg \dagger(z)$, $\dagger(z \div w) = \dagger(z) \div \dagger(w)$ (for $\div \in \{\wedge, \rightarrow\}$) and $\dagger(\perp_{\mathcal{A}}^T) = \perp_{\mathcal{A}}^P$. The details of the proof are left to the reader. $\square$

The last result shows that the algebras of triples $\mathbf{3}^X$ defined in §3 correspond to twist structures for **Ciore** over Boolean algebras of the form $\wp(X)$, as defined in Coniglio et al. (2018).

§6. QCiore-structures, satisfiability, and validity. Inspired by Proposition 3.2, the notion of **QCiore**-structure will be introduced in this section. As it will be shown below, these first-order structures will be adequate for characterizing the logic **QCiore**.

A *partial (or pragmatic) structure for QCiore over the signature Θ* is an ordered pair $\mathfrak{A} = \langle A, (\cdot)^{\mathfrak{A}} \rangle$ where $A \neq \emptyset$ and $(\cdot)^{\mathfrak{A}}$ is a function such that

- for all $R \in \mathcal{P}_n$, $R^{\mathfrak{A}} = \langle R_{\oplus}^{\mathfrak{A}}, R_{\ominus}^{\mathfrak{A}}, R_{\odot}^{\mathfrak{A}} \rangle$ is a triple over A^n ,
- $(\cdot)^{\mathfrak{A}}$ is defined as usual over $\mathcal{F}$ and $\mathcal{C}$.

If $\Theta = \langle \mathcal{P}, \mathcal{F}, \mathcal{C} \rangle$ and $\Theta' = \langle \mathcal{P}', \mathcal{F}', \mathcal{C}' \rangle$ are first-order signatures such that $\mathcal{P}'_n \subseteq \mathcal{P}_n$, $\mathcal{F}'_n \subseteq \mathcal{F}_n$ for every $n \geq 1$, and $\mathcal{C}' \subseteq \mathcal{C}$, we write $\Theta' \subseteq \Theta$. In this case, if $\mathfrak{A} = \langle A, (\cdot)^{\mathfrak{A}} \rangle$ is a partial structure for **QCiore** over Θ and $\mathfrak{A}' = \langle A, (\cdot)^{\mathfrak{A}'} \rangle$ is a partial structure for **QCiore** over Θ' such that $(\cdot)^{\mathfrak{A}'}$ is the restriction of the map $(\cdot)^{\mathfrak{A}}$ over Θ' , we say that $\mathfrak{A}'$ is the *reduct* of $\mathfrak{A}$ to Θ' , denoted by $\mathfrak{A}|_{\Theta'} = \mathfrak{A}'$.

Next, we recall some notions that are standard for Tarskian structures, as well as the notion of *quantification over sets* considered in Coniglio & Silvestrini (2014).

DEFINITION 6.1. Let A be a nonempty set. An assignment into A is any map $s : \mathcal{V} \rightarrow A$. We denote by $S(A)$ the set of all assignments into A , i.e., $S(A) = A^{\mathcal{V}}$. If $\mathfrak{A}$ is a partial structure over Θ with domain A then an assignment into $\mathfrak{A}$ is any assignment into A . The set of all assignments into $\mathfrak{A}$ will be denoted by $S(\mathfrak{A})$, i.e., $S(\mathfrak{A}) = S(A) = A^{\mathcal{V}}$.

DEFINITION 6.2. Let $s \in S(\mathfrak{A})$. The value of the term t in $\mathfrak{A}$ under the assignment s , denoted by $t^{\mathfrak{A}}[s]$, is defined inductively as follows:

- if t is v , for $v \in \mathcal{V}$, then $t^{\mathfrak{A}}[s] := s(v)$,
- if t is c , for $c \in \mathcal{C}$, then $t^{\mathfrak{A}}[s] := c^{\mathfrak{A}}$,
- if t is $f(t_1, \dots, t_n)$, for $f \in \mathcal{F}_n$ and terms t_i , then $t^{\mathfrak{A}}[s] := f^{\mathfrak{A}}(t_1^{\mathfrak{A}}[s], \dots, t_n^{\mathfrak{A}}[s])$.

DEFINITION 6.3. Let A be a nonempty set. Given $s \in S(A)$, $x \in \mathcal{V}$ and $a \in A$, the assignment $s_x^a : \mathcal{V} \rightarrow A$ is defined by

$$s_x^a(y) = \begin{cases} s(y) & \text{if } y \neq x \\ a & \text{if } y = x \end{cases} \quad (1)$$

DEFINITION 6.4. Let A be a nonempty set. Given a set Z let $\wp(Z)_+$ be the set of nonempty subsets of Z . For $x \in \mathcal{V}$ let $\widehat{\forall}x : \wp(S(A)) \rightarrow \wp(S(A))$ and $\widehat{\exists}x : \wp(S(A)) \rightarrow \wp(S(A))$ be defined as follows, for every $Y \subseteq S(A)$:

$$\widehat{\forall}x(Y) := \{s \in S(A) : s_x^a \in Y \text{ for all } a \in A\},$$

$$\widehat{\exists}x(Y) := \{s \in S(A) : s_x^a \in Y \text{ for some } a \in A\}.$$

The functions $\widetilde{\forall} : \wp(\mathbf{3})_+ \rightarrow \mathbf{3}$ and $\widetilde{\exists} : \wp(\mathbf{3})_+ \rightarrow \mathbf{3}$ are defined as follows, for every $\emptyset \neq Y \subseteq \mathbf{3}$:

$$\widetilde{\forall}(Y) := \begin{cases} 1 & \text{if } 1 \in Y, 0 \notin Y \\ \frac{1}{2} & \text{if } Y = \{\frac{1}{2}\} \\ 0 & \text{if } 0 \in Y \end{cases} \quad \widetilde{\exists}(Y) := \begin{cases} 1 & \text{if } Y \neq \{\frac{1}{2}\}, Y \neq \{0\} \\ \frac{1}{2} & \text{if } Y = \{\frac{1}{2}\} \\ 0 & \text{if } Y = \{0\} \end{cases}$$

DEFINITION 6.5. A **QCiore**-structure over Θ is a pair $\langle \mathfrak{A}, \|\cdot\|^{\mathfrak{A}} \rangle$ such that $\mathfrak{A}$ is a partial structure for **QCiore** and $\|\cdot\|^{\mathfrak{A}} : \mathcal{L}(\Theta) \rightarrow \mathbf{3}^{S(\mathfrak{A})}$ is a map defined recursively as follows, for every $s \in S(\mathfrak{A})$ (recall the operations defined in Remark 3.1 and Definition 6.4):

1. If $P(t_1, \dots, t_n)$ is atomic, $\|P(t_1, \dots, t_n)\|^{\mathfrak{A}}(s) = P^{\mathfrak{A}}(t_1^{\mathfrak{A}}[s], \dots, t_n^{\mathfrak{A}}[s])$,
2. $\|\varphi \wedge \psi\|^{\mathfrak{A}} = \|\varphi\|^{\mathfrak{A}} \wedge \|\psi\|^{\mathfrak{A}}$,
3. $\|\varphi \vee \psi\|^{\mathfrak{A}} = \|\varphi\|^{\mathfrak{A}} \vee \|\psi\|^{\mathfrak{A}}$,
4. $\|\varphi \rightarrow \psi\|^{\mathfrak{A}} = \|\varphi\|^{\mathfrak{A}} \rightarrow \|\psi\|^{\mathfrak{A}}$,
5. $\|\neg\varphi\|^{\mathfrak{A}} = \neg\|\varphi\|^{\mathfrak{A}}$,
6. $\|\circ\varphi\|^{\mathfrak{A}} = \circ\|\varphi\|^{\mathfrak{A}}$,
7. $\|\forall x\varphi\|^{\mathfrak{A}}(s) = \widetilde{\forall}(\{\|\varphi\|^{\mathfrak{A}}(s_x^a) : a \in A\})$,
8. $\|\exists x\varphi\|^{\mathfrak{A}}(s) = \widetilde{\exists}(\{\|\varphi\|^{\mathfrak{A}}(s_x^a) : a \in A\})$.

Observe that a partial structure for **QCiore** over Θ determines a unique **QCiore**-structure over Θ . Hence, both structures will be identified from now on.

PROPOSITION 6.6. *Let $\langle \mathfrak{A}, \|\cdot\|^{\mathfrak{A}} \rangle$ be a **QCiore**-structure over a signature Θ . Then, by denoting $\|\varphi\|^{\mathfrak{A}}$ as $\langle \|\varphi\|_{\oplus}^{\mathfrak{A}}, \|\varphi\|_{\ominus}^{\mathfrak{A}}, \|\varphi\|_{\odot}^{\mathfrak{A}} \rangle$ (as stipulated in §3), the following holds (recalling Remark 3.1):*

If $P(t_1, \dots, t_n)$ is atomic then, for every $\# \in \{\oplus, \ominus, \odot\}$ and $s \in S(\mathfrak{A})$:

$$s \in \|P(t_1, \dots, t_n)\|_{\#}^{\mathfrak{A}} \text{ iff } (t_1^{\mathfrak{A}}[s], \dots, t_n^{\mathfrak{A}}[s]) \in P_{\#}^{\mathfrak{A}},$$

$$\|\varphi \wedge \psi\|^{\mathfrak{A}} := \langle (\|\varphi\|_{\oplus}^{\mathfrak{A}} \cap \|\psi\|_{\oplus}^{\mathfrak{A}}) \cup (\|\varphi\|_{\oplus}^{\mathfrak{A}} \cap \|\psi\|_{\odot}^{\mathfrak{A}}) \cup (\|\varphi\|_{\odot}^{\mathfrak{A}} \cap \|\psi\|_{\oplus}^{\mathfrak{A}}), \\ \|\varphi\|_{\odot}^{\mathfrak{A}} \cup \|\psi\|_{\odot}^{\mathfrak{A}}, \|\varphi\|_{\odot}^{\mathfrak{A}} \cap \|\psi\|_{\odot}^{\mathfrak{A}} \rangle,$$

$$\|\varphi \vee \psi\|^{\mathfrak{A}} := \langle \|\varphi\|_{\oplus}^{\mathfrak{A}} \cup \|\psi\|_{\oplus}^{\mathfrak{A}} \cup (\|\varphi\|_{\odot}^{\mathfrak{A}} \cap \|\psi\|_{\odot}^{\mathfrak{A}}) \cup (\|\varphi\|_{\odot}^{\mathfrak{A}} \cup \|\psi\|_{\odot}^{\mathfrak{A}}), \\ \|\varphi\|_{\odot}^{\mathfrak{A}} \cap \|\psi\|_{\odot}^{\mathfrak{A}}, \|\varphi\|_{\odot}^{\mathfrak{A}} \cap \|\psi\|_{\odot}^{\mathfrak{A}} \rangle,$$

$$\|\varphi \rightarrow \psi\|^{\mathfrak{A}} := \langle \|\varphi\|_{\odot}^{\mathfrak{A}} \cup \|\psi\|_{\oplus}^{\mathfrak{A}} \cup (\|\varphi\|_{\oplus}^{\mathfrak{A}} \cap \|\psi\|_{\odot}^{\mathfrak{A}}), (\|\varphi\|_{\oplus}^{\mathfrak{A}} \cup \|\varphi\|_{\odot}^{\mathfrak{A}}) \cap \|\psi\|_{\odot}^{\mathfrak{A}}, \\ \|\varphi\|_{\odot}^{\mathfrak{A}} \cap \|\psi\|_{\odot}^{\mathfrak{A}} \rangle,$$

$$\|\neg\varphi\|^{\mathfrak{A}} := \langle \|\varphi\|_{\odot}^{\mathfrak{A}}, \|\varphi\|_{\oplus}^{\mathfrak{A}}, \|\varphi\|_{\odot}^{\mathfrak{A}} \rangle,$$

$$\|\circ\varphi\|^{\mathfrak{A}} := \langle \|\varphi\|_{\oplus}^{\mathfrak{A}} \cup \|\varphi\|_{\odot}^{\mathfrak{A}}, \|\varphi\|_{\odot}^{\mathfrak{A}}, \emptyset \rangle,$$

$$\|\forall x\varphi\|^{\mathfrak{A}} := \langle \widehat{\exists}x(\|\varphi\|_{\oplus}^{\mathfrak{A}}) - \widehat{\exists}x(\|\varphi\|_{\odot}^{\mathfrak{A}}), \widehat{\exists}x(\|\varphi\|_{\odot}^{\mathfrak{A}}), \widehat{\forall}x(\|\varphi\|_{\odot}^{\mathfrak{A}}) \rangle,$$

$$\|\exists x\varphi\|^{\mathfrak{A}} := \langle S(\mathfrak{A}) - (\widehat{\forall}x(\|\varphi\|_{\odot}^{\mathfrak{A}}) \cup \widehat{\forall}x(\|\varphi\|_{\oplus}^{\mathfrak{A}})), \widehat{\forall}x(\|\varphi\|_{\odot}^{\mathfrak{A}}), \widehat{\forall}x(\|\varphi\|_{\oplus}^{\mathfrak{A}}) \rangle.$$

Proof. Straightforward from the definitions and from Proposition 3.2. $\square$

REMARK 6.7. *Observe that, with notation as in Proposition 6.6: $\|\varphi\|_{\oplus}^{\mathfrak{A}} = \{s \in S(\mathfrak{A}) : \|\varphi\|^{\mathfrak{A}}(s) = 1\}$; $\|\varphi\|_{\odot}^{\mathfrak{A}} = \{s \in S(\mathfrak{A}) : \|\varphi\|^{\mathfrak{A}}(s) = \frac{1}{2}\}$; and $\|\varphi\|_{\ominus}^{\mathfrak{A}} = \{s \in S(\mathfrak{A}) : \|\varphi\|^{\mathfrak{A}}(s) = 0\}$, for every formula φ .*

REMARK 6.8. *Recall the structure of triples for **Ciore** introduced in Definition 5.2. By considering the Boolean algebra $\wp(S(\mathfrak{A}))$, Proposition 6.6 states operations $[\forall x]_T^{\mathfrak{A}}, [\exists x]_T^{\mathfrak{A}} : \mathcal{T}_{\wp(S(\mathfrak{A}))} \rightarrow \mathcal{T}_{\wp(S(\mathfrak{A}))}$ (for $x \in \mathcal{V}$) given by*

$$[\forall x]_T^{\mathfrak{A}} \langle Z_1, Z_2, Z_3 \rangle = \langle \widehat{\exists}x(Z_1) - \widehat{\exists}x(Z_2), \widehat{\exists}x(Z_2), \widehat{\forall}x(Z_3) \rangle,$$

$$[\exists x]_T^{\mathfrak{A}} \langle Z_1, Z_2, Z_3 \rangle = \langle S(\mathfrak{A}) - (\widehat{\forall}x(Z_2) \cup \widehat{\forall}x(Z_3)), \widehat{\forall}x(Z_2), \widehat{\forall}x(Z_3) \rangle.$$

By using the isomorphism $\dagger$ between $\mathcal{T}_A$ and $\mathcal{P}_A$ given in Proposition 5.4, as well as its inverse $\ddagger$, the operations $[\forall x]_P^{\mathfrak{A}}$ and $[\exists x]_P^{\mathfrak{A}}$ induce operations $[\forall x]_P^{\mathfrak{A}}, [\exists x]_P^{\mathfrak{A}} : \mathcal{P}_{\emptyset(S(\mathfrak{A}))} \rightarrow \mathcal{P}_{\emptyset(S(\mathfrak{A}))}$ given by $[\forall x]_P^{\mathfrak{A}} W := \dagger([\forall x]_T^{\mathfrak{A}}(\ddagger W))$ and $[\exists x]_P^{\mathfrak{A}} W := \dagger([\exists x]_T^{\mathfrak{A}}(\ddagger W))$, for $W = \langle W_1, W_2 \rangle$. Observe that $[\forall x]_P^{\mathfrak{A}} W$ and $[\exists x]_P^{\mathfrak{A}} W$ are given, respectively, by

$$\begin{aligned} & ((\widehat{\exists x}(W_1 - W_2) - \widehat{\exists x}(W_2 - W_1)) \cup \widehat{\forall x}(W_1 \cap W_2), \widehat{\exists x}(W_1 - W_2) \cup \widehat{\forall x}(W_1 \cap W_2)), \\ & ((\widehat{\forall x}(W_1 - W_2) - \widehat{\forall x}(W_2 - W_1)) \cup \widehat{\forall x}(W_1 \cap W_2), \widehat{\forall x}(W_1 - W_2) \cup \widehat{\forall x}(W_1 \cap W_2)). \end{aligned}$$

This shows how to define quantification over twist structures for **QCiore**, allowing to define twist structures semantics for **QCiore**. Observe that, by definition, $\dagger([\forall x]_T^{\mathfrak{A}} Z) = [\forall x]_P^{\mathfrak{A}}(\ddagger Z)$ and $\dagger([\exists x]_T^{\mathfrak{A}} Z) = [\exists x]_P^{\mathfrak{A}}(\ddagger Z)$ for every $Z = \langle Z_1, Z_2, Z_3 \rangle$. That is, the isomorphism $\dagger$ also preserves quantification.

DEFINITION 6.9. Let $\langle \mathfrak{A}, \|\cdot\|^{\mathfrak{A}} \rangle$ be a **QCiore**-structure, let φ be a formula and let $s \in S(\mathfrak{A})$. We say that s **QCiore**-satisfies φ in $\mathfrak{A}$, denoted by $\mathfrak{A} \Vdash_{\text{QCiore}} \varphi[s]$, if $s \in \|\varphi\|_{\oplus}^{\mathfrak{A}} \cup \|\varphi\|_{\odot}^{\mathfrak{A}}$. Besides, φ is said to be **QCiore**-valid in $\mathfrak{A}$, denoted by $\mathfrak{A} \Vdash_{\text{QCiore}} \varphi$, if $\|\varphi\|_{\oplus}^{\mathfrak{A}} \cup \|\varphi\|_{\odot}^{\mathfrak{A}} = S(\mathfrak{A})$. A **QCiore**-structure $\mathfrak{A}$ is a **QCiore**-model of a set Γ of formulas if $\mathfrak{A} \Vdash_{\text{QCiore}} \varphi$ for each $\varphi \in \Gamma$.

Given a set $\Gamma \cup \{\varphi\}$ of formulas, we say that φ is a **QCiore**-consequence of Γ , denoted by $\Gamma \vdash_{\text{QCiore}} \varphi$ if, for every **QCiore**-structure $\mathfrak{A}$, we have that: $\mathfrak{A} \Vdash_{\text{QCiore}} \psi$ for every $\psi \in \Gamma$ implies that $\mathfrak{A} \Vdash_{\text{QCiore}} \varphi$. A formula φ is **QCiore**-valid, denoted by $\Vdash_{\text{QCiore}} \varphi$, if $\emptyset \vdash_{\text{QCiore}} \varphi$.

The following propositions will be useful in the sequel.

PROPOSITION 6.10. Let $\langle \mathfrak{A}, \|\cdot\|^{\mathfrak{A}} \rangle$ be a **QCiore**-structure and let φ and ψ be formulas. Then

- (i) $\mathfrak{A} \Vdash_{\text{QCiore}} P(t_1, \dots, t_n)[s]$ iff $(t_1^{\mathfrak{A}}[s], \dots, t_n^{\mathfrak{A}}[s]) \in P_{\oplus}^{\mathfrak{A}} \cup P_{\odot}^{\mathfrak{A}}$, for every atomic formula $P(t_1, \dots, t_n)$.
- (ii) $\mathfrak{A} \Vdash_{\text{QCiore}} (\varphi \wedge \psi)[s]$ iff $\mathfrak{A} \Vdash_{\text{QCiore}} \varphi[s]$ and $\mathfrak{A} \Vdash_{\text{QCiore}} \psi[s]$.
- (iii) $\mathfrak{A} \Vdash_{\text{QCiore}} (\varphi \vee \psi)[s]$ iff $\mathfrak{A} \Vdash_{\text{QCiore}} \varphi[s]$ or $\mathfrak{A} \Vdash_{\text{QCiore}} \psi[s]$.
- (iv) $\mathfrak{A} \Vdash_{\text{QCiore}} (\varphi \rightarrow \psi)[s]$ iff $\mathfrak{A} \Vdash_{\text{QCiore}} \varphi[s]$ implies $\mathfrak{A} \Vdash_{\text{QCiore}} \psi[s]$.
- (v) $\mathfrak{A} \Vdash_{\text{QCiore}} \neg\varphi[s]$ iff $s \in \|\varphi\|_{\odot}^{\mathfrak{A}} \cup \|\varphi\|_{\oplus}^{\mathfrak{A}}$.
- (vi) $\mathfrak{A} \Vdash_{\text{QCiore}} \circ\varphi[s]$ iff $s \in \|\varphi\|_{\oplus}^{\mathfrak{A}} \cup \|\varphi\|_{\odot}^{\mathfrak{A}}$.
- (vii) $\mathfrak{A} \Vdash_{\text{QCiore}} \forall x\varphi[s]$ iff $\mathfrak{A} \Vdash_{\text{QCiore}} \varphi[s_x^a]$ for all $a \in A$.
- (viii) $\mathfrak{A} \Vdash_{\text{QCiore}} \exists x\varphi[s]$ iff there exists $a \in A$ such that $\mathfrak{A} \Vdash_{\text{QCiore}} \varphi[s_x^a]$.

Proof. It is consequence of Definition 6.5. □

PROPOSITION 6.11. Let φ a formula, $\mathfrak{A}$ a **QCiore**-structure and $s \in S(\mathfrak{A})$. Then:

- (i) $s \in \|\varphi\|_{\oplus}^{\mathfrak{A}}$ iff $\mathfrak{A} \Vdash_{\text{QCiore}} (\varphi \wedge \circ\varphi)[s]$.
- (ii) $s \in \|\varphi\|_{\odot}^{\mathfrak{A}}$ iff $\mathfrak{A} \Vdash_{\text{QCiore}} (\neg\varphi \wedge \circ\varphi)[s]$.
- (iii) $s \in \|\varphi\|_{\odot}^{\mathfrak{A}}$ iff $\mathfrak{A} \Vdash_{\text{QCiore}} (\varphi \wedge \neg\varphi)[s]$.
- (iv) s **QCiore**-satisfies in $\mathfrak{A}$ exactly two of the formulas φ , $\neg\varphi$, $\circ\varphi$.

Proof. (i) $\mathfrak{A} \Vdash_{\text{QCiore}} (\varphi \wedge \circ\varphi)[s]$ iff, by Proposition 6.10 (ii), $\mathfrak{A} \Vdash_{\text{QCiore}} \varphi[s]$ and $\mathfrak{A} \Vdash_{\text{QCiore}} \circ\varphi[s]$ iff, by Definition 6.9 and Proposition 6.10 (vi), $s \in \|\varphi\|_{\oplus}^{\mathfrak{A}} \cup \|\varphi\|_{\odot}^{\mathfrak{A}}$ and $s \in \|\varphi\|_{\oplus}^{\mathfrak{A}} \cup \|\varphi\|_{\odot}^{\mathfrak{A}}$ iff $s \in \|\varphi\|_{\oplus}^{\mathfrak{A}}$, since $\|\varphi\|_{\odot}^{\mathfrak{A}} \cap \|\varphi\|_{\oplus}^{\mathfrak{A}} = \emptyset$.

Items (ii), (iii), and (iv) are proved analogously. □

§7. Soundness and completeness. In this section, the soundness and completeness of **QCiore** with respect to its first-order structures will be proven. For this, it is necessary to establish some auxiliary results which will be used repeatedly in what follows. From now on $\langle \mathfrak{A}, \parallel \cdot \parallel^{\mathfrak{A}} \rangle$ denotes a **QCiore**-structure.

PROPOSITION 7.1. *The following statements are equivalent:*

- (i) $\mathfrak{A} \Vdash_{\mathbf{QCiore}} \varphi \rightarrow \psi$;
- (ii) $\|\varphi\|_{\oplus}^{\mathfrak{A}} \cup \|\varphi\|_{\odot}^{\mathfrak{A}} \subseteq \|\psi\|_{\oplus}^{\mathfrak{A}} \cup \|\psi\|_{\odot}^{\mathfrak{A}}$;
- (iii) $\|\psi\|_{\odot}^{\mathfrak{A}} \subseteq \|\varphi\|_{\odot}^{\mathfrak{A}}$.

Proof. It is consequence of Definition 6.5 and Proposition 6.10(iv). $\square$

Next we shall prove that the inference rules of **QCiore** preserve validity.

PROPOSITION 7.2. *If $\mathfrak{A} \Vdash_{\mathbf{QCiore}} \varphi$ and $\mathfrak{A} \Vdash_{\mathbf{QCiore}} \varphi \rightarrow \psi$, then $\mathfrak{A} \Vdash_{\mathbf{QCiore}} \psi$.*

Proof. From $\mathfrak{A} \Vdash_{\mathbf{QCiore}} \varphi$ it follows that $\|\varphi\|_{\oplus}^{\mathfrak{A}} \cup \|\varphi\|_{\odot}^{\mathfrak{A}} = S(\mathfrak{A})$. By Proposition 7.1 it follows that $\|\varphi\|_{\oplus}^{\mathfrak{A}} \cup \|\varphi\|_{\odot}^{\mathfrak{A}} \subseteq \|\psi\|_{\oplus}^{\mathfrak{A}} \cup \|\psi\|_{\odot}^{\mathfrak{A}}$. Hence, $\mathfrak{A} \Vdash_{\mathbf{QCiore}} \psi$. $\square$

Let Θ be a first-order signature. An *instance* of a propositional formula $\varphi = \varphi(p_1, \dots, p_n)$ in $\mathcal{L}$ is a formula in $\mathcal{L}(\Theta)$ obtained from φ by simultaneously substituting each occurrence of the propositional variable p_i by a formula β_i in $\mathcal{L}(\Theta)$, for $1 \leq i \leq n$. We denote such instance by $\varphi[\beta_1/p_1, \dots, \beta_n/p_n]$.

PROPOSITION 7.3. *Let $\alpha = \varphi[\beta_1/p_1, \dots, \beta_n/p_n]$ be an instance in $\mathcal{L}(\Theta)$ of a propositional formula $\varphi(p_1, \dots, p_n)$ in $\mathcal{L}$. For every $s \in S(\mathfrak{A})$ we define the valuation v_s over the 3-valued matrix M_e for **Ciore** (recall Theorem 2.1) as follows:*

$$v_s(p_i) = \begin{cases} 1 & \text{if } s \in \|\beta_i\|_{\oplus}^{\mathfrak{A}} \\ \frac{1}{2} & \text{if } s \in \|\beta_i\|_{\odot}^{\mathfrak{A}} \\ 0 & \text{if } s \in \|\beta_i\|_{\ominus}^{\mathfrak{A}} \end{cases}$$

for $1 \leq i \leq n$, and $v_s(p_i)$ is arbitrary otherwise. Then $\|\alpha\|^{\mathfrak{A}}(s) = v_s(\varphi)$.

Proof. Straightforward by induction on the complexity of φ , taking into account Definition 6.5 and Remark 6.7. $\square$

As a direct consequence, we have the following

COROLLARY 7.4. *If φ is a instance of a **Ciore**-tautology, then $\mathfrak{A} \Vdash_{\mathbf{QCiore}} \varphi$.*

PROPOSITION 7.5. *Let $t(x_1, \dots, x_n)$ be a term and let $\varphi(x_1, \dots, x_n)$ be a formula in $\mathcal{L}(\Theta)$. If s and s' are assignments in $\mathfrak{A}$ such that $s(x_i) = s'(x_i)$ for all i ($1 \leq i \leq n$), then (i) $t^{\mathfrak{A}}[s] = t^{\mathfrak{A}}[s']$, (ii) $\|\varphi\|^{\mathfrak{A}}(s) = \|\varphi\|^{\mathfrak{A}}(s')$, (iii) $\mathfrak{A} \Vdash_{\mathbf{QCiore}} \varphi[s]$ iff $\mathfrak{A} \Vdash_{\mathbf{QCiore}} \varphi[s']$.*

Proof. Straightforward, by induction on the length of the term t and the complexity of φ . $\square$

THEOREM 7.6. *Let φ be a sentence. Then: either $\|\varphi\|_{\oplus}^{\mathfrak{A}} = S(\mathfrak{A})$ or $\|\varphi\|_{\odot}^{\mathfrak{A}} = S(\mathfrak{A})$ or $\|\varphi\|_{\ominus}^{\mathfrak{A}} = S(\mathfrak{A})$.*

Proof. Let φ be a sentence and let $s \in S(\mathfrak{A})$. Suppose that $s \in \|\varphi\|_{\oplus}^{\mathfrak{A}}$. Then, for all $s' \in S(\mathfrak{A})$ it holds vacuously that $s(x) = s'(x)$ for every variable x occurring free in φ . Then

$s' \in \|\varphi\|_{\oplus}^{\mathfrak{A}}$, by Proposition 7.5, and so $\|\varphi\|_{\oplus}^{\mathfrak{A}} = S(\mathfrak{A})$. Analogously, $s \in \|\varphi\|_{\ominus}^{\mathfrak{A}}$ implies that $\|\varphi\|_{\ominus}^{\mathfrak{A}} = S(\mathfrak{A})$, and $s \in \|\varphi\|_{\odot}^{\mathfrak{A}}$ implies that $\|\varphi\|_{\odot}^{\mathfrak{A}} = S(\mathfrak{A})$. $\square$

REMARK 7.7. *Observe that if φ is a sentence then there exists $\mathbf{v} \in \mathbf{3}$ such that $\|\varphi\|^{\mathfrak{A}}(s) = \mathbf{v}$ for all $s \in S(\mathfrak{A})$. In this case, we shall write $\|\varphi\|^{\mathfrak{A}} = \mathbf{v}$.*

COROLLARY 7.8. *If φ is a sentence then one and only one of the following statements holds:*

- (i) $\mathfrak{A} \Vdash_{\mathbf{QCiore}} \varphi \wedge \circ\varphi$,
- (ii) $\mathfrak{A} \Vdash_{\mathbf{QCiore}} \neg\varphi \wedge \circ\varphi$,
- (iii) $\mathfrak{A} \Vdash_{\mathbf{QCiore}} \varphi \wedge \neg\varphi$.

Let t and u be terms and $s \in S(\mathfrak{A})$. Let t' be the term that results from t by replacing simultaneously all the occurrences of the variable x by u , which will be denoted by $t(u/x)$. Let s_x^a the assignment defined as in Definition 6.3, where $a = u^{\mathfrak{A}}[s]$ (recall Definition 6.2). It is easy to prove, by induction on the complexity of the term t , that $(t')^{\mathfrak{A}}[s] = t^{\mathfrak{A}}[s_x^a]$. Moreover:

PROPOSITION 7.9. *Let t be a term free for x in φ , and let $a = t^{\mathfrak{A}}[s]$. Then:*

- (i) $\|\varphi(t/x)\|^{\mathfrak{A}}(s) = \|\varphi\|^{\mathfrak{A}}(s_x^a)$ (Substitution Lemma),
- (ii) $\mathfrak{A} \Vdash_{\mathbf{QCiore}} \varphi(t/x)[s]$ iff $\mathfrak{A} \Vdash_{\mathbf{QCiore}} \varphi[s_x^a]$,
- (iii) If $\mathfrak{A} \Vdash_{\mathbf{QCiore}} \forall x\varphi[s]$, then $\mathfrak{A} \Vdash_{\mathbf{QCiore}} \varphi(t/x)[s]$.

Proof. It is routine. $\square$

PROPOSITION 7.10. (i) *Let t be a term free for x in φ . Then:*

- (a) $\mathfrak{A} \Vdash_{\mathbf{QCiore}} \forall x\varphi \rightarrow \varphi(t/x)$, (b) $\mathfrak{A} \Vdash_{\mathbf{QCiore}} \varphi(t/x) \rightarrow \exists x\varphi$.

- (ii) $\mathfrak{A} \Vdash_{\mathbf{QCiore}} \exists x\circ\varphi \leftrightarrow \circ\forall x\varphi$,
- (iii) $\mathfrak{A} \Vdash_{\mathbf{QCiore}} \exists x\circ\varphi \leftrightarrow \circ\exists x\varphi$.

Proof.

- (i) They are consequence of Lemma 7.9.
- (ii) It follows from $\|\circ\forall x\varphi\|_{\ominus}^{\mathfrak{A}} = \|\forall x\varphi\|_{\odot}^{\mathfrak{A}} = \widehat{\forall x}(\|\varphi\|_{\odot}^{\mathfrak{A}}) = \widehat{\forall x}(\|\circ\varphi\|_{\ominus}^{\mathfrak{A}}) = \|\exists x\circ\varphi\|_{\ominus}^{\mathfrak{A}}$.
- (iii) It follows from $\|\exists x\circ\varphi\|_{\ominus}^{\mathfrak{A}} = \widehat{\forall x}(\|\circ\varphi\|_{\ominus}^{\mathfrak{A}}) = \widehat{\forall x}(\|\varphi\|_{\odot}^{\mathfrak{A}}) = \|\exists x\varphi\|_{\odot}^{\mathfrak{A}} = \|\circ\exists x\varphi\|_{\ominus}^{\mathfrak{A}}$. $\square$

PROPOSITION 7.11. (i) $\varphi \rightarrow \psi \Vdash_{\mathbf{QCiore}} \varphi \rightarrow \forall x\psi$ if the variable x does not occur free in φ .

- (ii) $\varphi \rightarrow \psi \Vdash_{\mathbf{QCiore}} \exists x\varphi \rightarrow \psi$ if the variable x does not occur free in ψ .

Proof. Let $\mathfrak{A}$ be a **QCiore**-structure and suppose that $\mathfrak{A} \Vdash_{\mathbf{QCiore}} \varphi \rightarrow \psi$. Then, by Proposition 7.1, $\|\psi\|_{\ominus}^{\mathfrak{A}} \subseteq \|\varphi\|_{\ominus}^{\mathfrak{A}}$.

(i) Since x does not occur free in φ and from Proposition 7.5, it follows that $\widehat{\exists x}(\|\varphi\|_{\ominus}^{\mathfrak{A}}) = \|\varphi\|_{\ominus}^{\mathfrak{A}}$. Since $\widehat{\exists x}(\|\psi\|_{\ominus}^{\mathfrak{A}}) \subseteq \widehat{\exists x}(\|\varphi\|_{\ominus}^{\mathfrak{A}})$, it follows that $\|\forall x\psi\|_{\odot}^{\mathfrak{A}} \subseteq \|\varphi\|_{\odot}^{\mathfrak{A}}$. Then, $\mathfrak{A} \Vdash_{\mathbf{QCiore}} \varphi \rightarrow \forall x\psi$, by Proposition 7.1.

(ii) Since x does not occur free in ψ and from Proposition 7.5, it follows that $\widehat{\forall x}(\|\psi\|_{\ominus}^{\mathfrak{A}}) =$

$\|\psi\|_{\Theta}^{\mathfrak{A}}$. Since $\widehat{\forall}x(\|\psi\|_{\Theta}^{\mathfrak{A}}) \subseteq \widehat{\forall}x(\|\varphi\|_{\Theta}^{\mathfrak{A}})$, it follows that $\|\psi\|_{\Theta}^{\mathfrak{A}} \subseteq \|\exists x\varphi\|_{\Theta}^{\mathfrak{A}}$. Then, $\mathfrak{A} \Vdash_{\mathbf{QCiore}} \exists x\varphi \rightarrow \psi$, by Proposition 7.1. $\square$

From the results stated in this section, we have:

THEOREM 7.12 (Soundness). *Let $\Gamma \cup \{\varphi\}$ be a set of formulas over Γ . If $\Gamma \vdash_{\mathbf{QCiore}} \varphi$ then $\Gamma \Vdash_{\mathbf{QCiore}} \varphi$.*

Proof. By induction on the complexity of φ and taking into account that all the axioms of **QCiore** are valid (see Corollary 7.4 and Proposition 7.10) and that all the inference rules of **QCiore** preserve validity in a given **QCiore**-structure (see Propositions 7.2 and 7.11). $\square$

In order to prove completeness, some well-known notions and results from classical first-order logic will be adapted to this framework. Recall that a *theory* is any set of sentences (that is, a set of formulas without free variables). A theory $T \subseteq \text{Sent}(\Theta)$ is **QCiore-consistent** if for every $\varphi \in \text{Sent}(\Theta)$ at most two of φ , $\neg\varphi$, $\circ\varphi$ are derivable from T in **QCiore** (recalling from §4 that $\text{Sent}(\Theta)$ denotes the set of sentences over Θ).

PROPOSITION 7.13. *A theory T is not **QCiore-consistent** if and only if $T \vdash_{\mathbf{QCiore}} \varphi$ for every sentence φ .*

Proof. Immediate, taking into account axiom (**bc1**). $\square$

COROLLARY 7.14. *T is **QCiore-consistent** if and only if there exists a sentence that is not derivable from T in **QCiore**.*

Proof. It is a consequence of Proposition 7.13. $\square$

PROPOSITION 7.15. *Let T be a theory and let φ be a sentence.*

- (i) φ is not derivable from T in **QCiore** iff $T \cup \{\neg\varphi, \circ\varphi\}$ is **QCiore-consistent**,
- (ii) $\neg\varphi$ is not derivable from T in **QCiore** iff $T \cup \{\varphi, \circ\varphi\}$ is **QCiore-consistent**,
- (iii) $\circ\varphi$ is not derivable from T in **QCiore** iff $T \cup \{\varphi, \neg\varphi\}$ is **QCiore-consistent**.

Proof. It is a consequence, respectively, of the following schemas of **Ciore**-tautologies (which are therefore theorems of **QCiore**): $((\neg\varphi \wedge \circ\varphi) \rightarrow \varphi) \rightarrow \varphi$, $((\varphi \wedge \circ\varphi) \rightarrow \neg\varphi) \rightarrow \neg\varphi$ and $((\varphi \wedge \neg\varphi) \rightarrow \circ\varphi) \rightarrow \circ\varphi$. $\square$

We shall say that a theory T is *maximal **QCiore-consistent*** if T is **QCiore-consistent** and the only **QCiore-consistent** theory which contains T is T itself.

PROPOSITION 7.16. *Let T be a maximal **QCiore-consistent** theory. Then, for all sentences φ and ψ :*

- (i) $\varphi \in T$ iff $T \vdash_{\mathbf{QCiore}} \varphi$;
- (ii) $\varphi \notin T$ iff $\{\neg\varphi, \circ\varphi\} \subseteq T$;
- (iii) $\neg\varphi \notin T$ iff $\{\varphi, \circ\varphi\} \subseteq T$;
- (iv) $\circ\varphi \notin T$ iff $\{\varphi, \neg\varphi\} \subseteq T$;
- (v) $\varphi \wedge \psi \in T$ iff $\varphi \in T$ and $\psi \in T$;
- (vi) $\varphi \vee \psi \in T$ iff $\varphi \in T$ or $\psi \in T$;
- (vii) exactly two of φ , $\neg\varphi$, $\circ\varphi$ are in T ;
- (viii) if $\forall x\varphi \in T$, then $\varphi(c/x) \in T$ for every $c \in \mathcal{C}$.

Proof. (i)

$$\begin{aligned}
 T \vdash_{\mathbf{QCiore}} \varphi & \\
 \implies T \not\vdash_{\mathbf{QCiore}} \neg\varphi \text{ or } T \not\vdash_{\mathbf{QCiore}} \circ\varphi & \quad (\text{by } \mathbf{QCiore}\text{-consistency}) \\
 \implies T \cup \{\varphi, \circ\varphi\} \text{ is } \mathbf{QCiore}\text{-consistent} & \\
 \text{or } T \cup \{\varphi, \neg\varphi\} \text{ is } \mathbf{QCiore}\text{-consistent} & \quad (\text{by Proposition 7.15(ii)(iii)}) \\
 \implies \varphi \in T & \quad (\text{by maximality}).
 \end{aligned}$$

(ii), (iii), and (iv): They are consequence of Proposition 7.15.

(v) It follows from **(Ax4)**, **(Ax5)** and **(Ax3)**.

(vi) ($\Rightarrow$) Assume that $\varphi \vee \psi \in T$. If $\varphi \notin T$ then $(\neg\varphi \wedge \circ\varphi) \wedge (\varphi \vee \psi) \in T$, from items (ii) and (v). Since $\vdash_{\mathbf{QCiore}} ((\neg\varphi \wedge \circ\varphi) \wedge (\varphi \vee \psi)) \rightarrow \psi$, it follows from item (i) that $\psi \in T$.

($\Leftarrow$) It is a consequence of **(Ax6)** and **(Ax7)**.

(vii) It follows from items (ii), (iii), (iv), and (v).

(viii) It is a consequence of **(Ax12)**. $\square$

Given that **QCiore** is a Tarskian and finitary logic, the Lindenbaum-Łoś Lemma (see (Wójcicki, 1984, Theorem 22.2) or (Carnielli & Coniglio, 2016, Theorem 2.2.6)) holds for this logic. In particular, this result holds when **QCiore** is restricted to sentences. Moreover, because of Corollary 7.14, the Lindenbaum-Łoś Lemma for the logic **QCiore** restricted to sentences is equivalent to the following:

THEOREM 7.17 (Lindenbaum-Łoś Lemma). *Any **QCiore**-consistent theory can be extended to a maximal **QCiore**-consistent theory.*

A set of sentences T is a **QCiore**-Henkin theory iff for any formula φ with at most one free variable x there is a constant c such that $\exists x\varphi \rightarrow \varphi(c/x) \in T$. Then, by a simple adaptation of the well-known method of constants introduced by Henkin (also using the weak version of the deduction metatheorem, that is, Theorem 4.4 restricted to sentences) we can prove the following:

THEOREM 7.18. *Every **QCiore**-consistent theory T over a signature Θ has a maximal **QCiore**-consistent and **QCiore**-Henkin extension over a signature Θ_C obtained from Θ by adding a set C of new individual constants.*

THEOREM 7.19 (Henkin Lemma). *Let T be a maximal **QCiore**-consistent and **QCiore**-Henkin theory. Then T has a **QCiore**-model.*

Proof. Let us define an **QCiore**-structure $\mathfrak{A}$ over Θ as follows:

- the domain A is the set of all closed terms over Θ ,
- if c is a constant symbol of Θ then $c^{\mathfrak{A}} := c$,
- if f is a n -ary function symbol of Θ , $f^{\mathfrak{A}} : A^n \rightarrow A$ is given by $f^{\mathfrak{A}}(a_1, \dots, a_n) := f(a_1, \dots, a_n)$,
- if R is a n -ary predicate symbol, $R^{\mathfrak{A}} = \langle R_{\oplus}^{\mathfrak{A}}, R_{\ominus}^{\mathfrak{A}}, R_{\odot}^{\mathfrak{A}} \rangle$ is defined as follows

$$\begin{aligned}
 (a_1, \dots, a_n) \in R_{\oplus}^{\mathfrak{A}} & \quad \text{iff} \quad R(a_1, \dots, a_n) \wedge \circ R(a_1, \dots, a_n) \in T \\
 (a_1, \dots, a_n) \in R_{\ominus}^{\mathfrak{A}} & \quad \text{iff} \quad \neg R(a_1, \dots, a_n) \wedge \circ R(a_1, \dots, a_n) \in T \\
 (a_1, \dots, a_n) \in R_{\odot}^{\mathfrak{A}} & \quad \text{iff} \quad R(a_1, \dots, a_n) \wedge \neg R(a_1, \dots, a_n) \in T
 \end{aligned}$$

By induction on the complexity of a sentence φ in in $\text{Sent}(\Theta)$ it will be proven that

- | | | | |
|-----|--|-----|---|
| (1) | $\ \varphi\ ^{\mathfrak{A}} = 1$ | iff | $\varphi \wedge \circ\varphi \in T$ |
| (2) | $\ \varphi\ ^{\mathfrak{A}} = 0$ | iff | $\neg\varphi \wedge \circ\varphi \in T$ |
| (3) | $\ \varphi\ ^{\mathfrak{A}} = \frac{1}{2}$ | iff | $\varphi \wedge \neg\varphi \in T$. |

Base step: Let φ be the atomic formula $R(a_1, \dots, a_n)$. Then, $\|R(a_1, \dots, a_n)\|^{\mathfrak{A}} = 1$ iff $(a_1, \dots, a_n) \in R_{\oplus}^{\mathfrak{A}}$ iff (by definition) $R(a_1, \dots, a_n) \wedge \circ R(a_1, \dots, a_n) \in T$. Hence, $\mathfrak{A}$ fulfills Condition (1). Similarly, we prove Conditions (2) and (3).

(IH): Suppose that the theorem holds for every formula ψ whose complexity is less or equal than n .

Observe that, in the inductive step (that is, when analyzing a formula φ with complexity $n + 1$), it is enough to prove the ‘only if’ direction of (1), (2), and (3), because of Proposition 7.16. Indeed, assuming that the ‘only if’ direction of (1), (2), and (3) was proved by a given φ , the corresponding ‘if’ direction of (1), (2), and (3) can be proven as follows: if $\|\varphi\|^{\mathfrak{A}} = 1$ then $\|\varphi\|^{\mathfrak{A}} \neq 0$ and $\|\varphi\|^{\mathfrak{A}} \neq \frac{1}{2}$. Then, by the ‘only if’ part of (2) and (3), $\varphi \wedge \neg\varphi \notin T$ and $\neg\varphi \wedge \circ\varphi \notin T$. By items (v) and (vii) of Proposition 7.16, $\varphi \wedge \circ\varphi \in T$. The cases $\|\varphi\|^{\mathfrak{A}} = 0$ and $\|\varphi\|^{\mathfrak{A}} = \frac{1}{2}$ are proved analogously.

Thus, in order to complete the proof let us prove, assuming (IH), the ‘only if’ part of (1), (2), and (3) for a given formula φ with complexity $n + 1$. We have the following cases to consider:

Case a. φ is $\neg\psi$

(a.1) If $\neg\psi \wedge \circ\neg\psi \in T$ then, by Theorem 2.2(vi), $\neg\psi \wedge \circ\psi \in T$ and so, by (HI), $\|\psi\|^{\mathfrak{A}} = 0$. From this $\|\neg\psi\|^{\mathfrak{A}} = \neg\|\psi\|^{\mathfrak{A}} = 1$.

(a.2) If $\neg\neg\psi \wedge \circ\neg\psi \in T$ then, by (cf) and Theorem 2.2(vi), $\psi \wedge \circ\psi \in T$ and so, by (HI), $\|\psi\|^{\mathfrak{A}} = 1$. Then $\|\neg\psi\|^{\mathfrak{A}} = \neg\|\psi\|^{\mathfrak{A}} = 0$.

(a.3) If $\neg\psi \wedge \neg\neg\psi \in T$ then, by (cf), $\psi \wedge \neg\psi \in T$ and so, by (HI), $\|\psi\|^{\mathfrak{A}} = \frac{1}{2}$. From this $\|\neg\psi\|^{\mathfrak{A}} = \neg\|\psi\|^{\mathfrak{A}} = \frac{1}{2}$.

Case b. φ is $\circ\psi$

(b.1) If $\circ\psi \wedge \circ\circ\psi \in T$ then $\circ\psi \in T$ and so, by (HI) and Proposition 7.16, $\|\psi\|^{\mathfrak{A}} \in \{1, 0\}$. Then $\|\circ\psi\|^{\mathfrak{A}} = \circ\|\psi\|^{\mathfrak{A}} = 1$.

(b.2) If $\neg\circ\psi \wedge \circ\circ\psi \in T$ then $\neg\circ\psi \in T$. By Theorem 2.2(iv), $\psi \wedge \neg\psi \in T$ and so, by (HI), $\|\psi\|^{\mathfrak{A}} = \frac{1}{2}$. This implies that $\|\circ\psi\|^{\mathfrak{A}} = \circ\|\psi\|^{\mathfrak{A}} = 0$.

(b.3) Since T is **QCiore**-consistent and $\circ\circ\psi \in T$ (by Theorem 2.2(v)) it follows that $\circ\psi \wedge \neg\circ\psi \notin T$. Then, this subcase holds vacuously.

Case c. φ is $\psi \wedge \xi$

(c.1) If $(\psi \wedge \xi) \wedge \circ(\psi \wedge \xi) \in T$ then, by Theorem 2.2(ix), $((\psi \wedge (\xi \wedge \circ\xi)) \vee ((\psi \wedge \circ\psi) \wedge \xi)) \in T$. Hence, by (IH) and Proposition 7.16: either $\|\psi\|^{\mathfrak{A}} \in \{1, \frac{1}{2}\}$ and $\|\xi\|^{\mathfrak{A}} = 1$, or $\|\psi\|^{\mathfrak{A}} = 1$ and $\|\xi\|^{\mathfrak{A}} \in \{1, \frac{1}{2}\}$. From this $\|\psi \wedge \xi\|^{\mathfrak{A}} = 1$.

(c.2) If $\neg(\psi \wedge \xi) \wedge \circ(\psi \wedge \xi) \in T$ then $(\neg\psi \wedge \circ\psi) \vee (\neg\xi \wedge \circ\xi) \in T$, by Theorem 2.2(vii). Hence, by (IH) and Proposition 7.16: either $\|\psi\|^{\mathfrak{A}} = 0$ or $\|\xi\|^{\mathfrak{A}} = 0$. From this $\|\psi \wedge \xi\|^{\mathfrak{A}} = \|\psi\|^{\mathfrak{A}} \wedge \|\xi\|^{\mathfrak{A}} = 0$.

(c.3) If $(\psi \wedge \xi) \wedge \neg(\psi \wedge \xi) \in T$ then $(\psi \wedge \neg\psi) \wedge (\xi \wedge \neg\xi) \in T$, by Theorem 2.2(viii). By (IH) and Proposition 7.16 it follows that $\|\psi\|^{\mathfrak{A}} = \|\xi\|^{\mathfrak{A}} = \frac{1}{2}$ and so $\|\psi \wedge \xi\|^{\mathfrak{A}} = \|\psi\|^{\mathfrak{A}} \wedge \|\xi\|^{\mathfrak{A}} = \frac{1}{2}$.

Case d. φ is $\psi \vee \xi$

The three subcases are treated as in **Case c**, but now using Theorems 2.2(x), 2.2(xi), and 2.2(xii), respectively.

Case e. φ is $\psi \rightarrow \xi$

The three subcases are treated as in **Case c**, but now using Theorems 2.2(xiii), 2.2(xiv), and 2.2(xv), respectively.

Case f. φ is $\exists x\psi$

(f.1) Suppose that $\exists x\psi \wedge \circ\exists x\psi \in T$. Since $\exists x\psi \in T$ and T is **QCiore**-Henkin, there exists a constant c such that $\psi(c/x) \in T$. By (IH), we have that $\|\psi(c/x)\|^{\mathfrak{A}} \in \{1, \frac{1}{2}\}$. Then, $\widehat{\nabla x}(\|\psi\|_{\oplus}^{\mathfrak{A}}) = \emptyset$. Since $\circ\exists x\psi \in T$ then $\exists x\circ\psi \in T$, by (Ax13). Using once again that T is a **QCiore**-Henkin theory, we conclude that there exists a witness d such that $\circ\psi(d/x) \in T$. By (IH), we have that $\|\psi(d/x)\|^{\mathfrak{A}} \in \{1, 0\}$. Then, $\widehat{\nabla x}(\|\psi\|_{\oplus}^{\mathfrak{A}}) = \emptyset$ and so $\|\exists x\psi\|_{\oplus}^{\mathfrak{A}} = S(\mathfrak{A})$. From this $\|\exists x\psi\|^{\mathfrak{A}} = 1$.

(f.2) If $\exists x\psi \wedge \neg\exists x\psi \in T$ then, by Theorem 4.2(iii), $\forall x(\psi \wedge \neg\psi) \in T$. By (Ax12) we have that $\psi(c/x) \wedge \neg\psi(c/x) \in T$, for all $c \in A$. Then, by (IH), $\|\psi(c/x)\|^{\mathfrak{A}} = S(\mathfrak{A})$ for all $c \in A$ and therefore $\|\exists x\psi\|_{\oplus}^{\mathfrak{A}} = \widehat{\nabla x}(\|\psi\|_{\oplus}^{\mathfrak{A}}) = S(\mathfrak{A})$. From this $\|\exists x\psi\|^{\mathfrak{A}} = \frac{1}{2}$.

(f.3) If $\neg\exists x\psi \wedge \circ\exists x\psi \in T$ then, by Theorem 4.2(i), $\forall x(\neg\psi \wedge \circ\psi) \in T$. By (Ax12) we have that $\neg\psi(c/x) \wedge \circ\psi(c/x) \in T$, for all $c \in A$. By (IH), $\|\psi(c/x)\|^{\mathfrak{A}} = S(\mathfrak{A})$ for all $c \in A$ and then $\|\exists x\psi\|_{\oplus}^{\mathfrak{A}} = \widehat{\nabla x}(\|\psi\|_{\oplus}^{\mathfrak{A}}) = S(\mathfrak{A})$. From this $\|\exists x\psi\|^{\mathfrak{A}} = 0$.

Case g. φ is $\forall x\psi$

(g.1) Suppose that $\forall x\psi \wedge \circ\forall x\psi \in T$. Since $\vdash_{\text{QCiore}} (\forall x\psi \wedge \circ\forall x\psi) \rightarrow (\exists x\psi \wedge \circ\exists x\psi)$ then $\exists x\psi \wedge \circ\exists x\psi \in T$. Since T is **QCiore**-Henkin, there exists a witness c such that $\psi(c/x) \wedge \circ\psi(c/x) \in T$. By (IH), it follows that $\|\psi(c/x)\|^{\mathfrak{A}} = 1$ and so $\|\psi(c/x)\|_{\oplus}^{\mathfrak{A}} = S(\mathfrak{A})$. Then $\widehat{\nabla x}(\|\psi\|_{\oplus}^{\mathfrak{A}}) = S(\mathfrak{A})$.

Suppose now that there exists $s \in \widehat{\nabla x}(\|\psi(x)\|_{\oplus}^{\mathfrak{A}})$. By (IH), there exists d such that $\neg\psi(d/x) \wedge \circ\psi(d/x) \in T$. Since $\forall x\psi \in T$ we conclude that $\psi(d/x) \wedge \neg\psi(d/x) \wedge \circ\psi(d/x) \in T$, which contradicts the fact that T is **QCiore**-consistent. Therefore $\widehat{\nabla x}(\|\psi\|_{\oplus}^{\mathfrak{A}}) = \emptyset$ and then $\|\forall x\psi\|_{\oplus}^{\mathfrak{A}} = \widehat{\nabla x}(\|\psi\|_{\oplus}^{\mathfrak{A}}) - \widehat{\nabla x}(\|\psi\|_{\oplus}^{\mathfrak{A}}) = S(\mathfrak{A})$. From this $\|\forall x\psi\|^{\mathfrak{A}} = 1$.

(g.2) If $\neg\forall x\psi \wedge \circ\forall x\psi \in T$ then $\exists x(\neg\psi \wedge \circ\psi) \in T$, by Theorem 4.2(ii). Since T is **QCiore**-Henkin, there exists c such that $\neg\psi(c/x) \wedge \circ\psi(c/x) \in T$. By (IH), $\|\psi(c/x)\|^{\mathfrak{A}} = 0$ and so $\|\psi(c/x)\|_{\oplus}^{\mathfrak{A}} = S(\mathfrak{A})$. Then, $\|\forall x\psi\|_{\oplus}^{\mathfrak{A}} = \widehat{\nabla x}(\|\psi\|_{\oplus}^{\mathfrak{A}}) = S(\mathfrak{A})$. From this $\|\forall x\psi\|^{\mathfrak{A}} = 0$.

(g.3) If $\forall x\psi \wedge \neg\forall x\psi \in T$ then $\forall x(\psi \wedge \neg\psi) \in T$, by Theorem 4.2(iv). Hence $\psi(c/x) \wedge \neg\psi(c/x) \in T$ for every $c \in A$. By (IH), we conclude that $\|\psi(c/x)\|^{\mathfrak{A}} = S(\mathfrak{A})$ for all $c \in A$. Then, $\|\forall x\psi\|_{\oplus}^{\mathfrak{A}} = \widehat{\nabla x}(\|\psi\|_{\oplus}^{\mathfrak{A}}) = S(\mathfrak{A})$. From this $\|\forall x\psi\|^{\mathfrak{A}} = \frac{1}{2}$. $\square$

THEOREM 7.20 (Completeness for sentences). *Let $\Gamma \cup \{\varphi\}$ be a set of sentences over Θ . Then $\Gamma \Vdash_{\text{QCiore}} \varphi$ implies $\Gamma \vdash_{\text{QCiore}} \varphi$.*

Proof. Suppose that φ is not derivable from Γ in **QCiore**. Then, by Proposition 7.15, we have that the theory $\Gamma \cup \{\neg\varphi, \circ\varphi\}$ is **QCiore**-consistent. By Theorem 7.18, there is a theory T over a signature Θ_C (obtained from Θ by adding a set C of new constants) which

is maximal **QCiore**-consistent and **QCiore**-Henkin, and such that $\Gamma \cup \{\neg\varphi, \circ\varphi\} \subseteq T$. By Theorem 7.19, there exists an **QCiore**-model $\mathfrak{A}$ over Θ_C for T . Let $\mathfrak{A}'$ be the reduct of $\mathfrak{A}$ to Θ . Then, $\mathfrak{A}' \models_{\mathbf{QCiore}} \Gamma$ and $\mathfrak{A}' \models_{\mathbf{QCiore}} \neg\varphi \wedge \circ\varphi$, and so $\mathfrak{A}' \not\models_{\mathbf{QCiore}} \varphi$. Therefore, we have that $\Gamma \not\models_{\mathbf{QCiore}} \varphi$. $\square$

REMARK 7.21. *Observe that completeness was proved only for sentences, while soundness was stated for formulas in general (recall Theorem 7.12). However, a completeness theorem for formulas in general (i.e., for formulas possibly having free variables) can be obtained from Theorem 7.20 by observing the following: for any formula ψ let $(\forall)\psi$ be the universal closure of ψ , that is: if ψ is a sentence then $(\forall)\psi = \psi$, and if ψ has exactly the variables $x_1, \dots, x_n$ occurring free then $(\forall)\psi = (\forall x_1) \dots (\forall x_n)\psi$. If Γ is a set of formulas then $(\forall)\Gamma := \{(\forall)\psi : \psi \in \Gamma\}$. Thus, it is easy to see that, for every set $\Gamma \cup \{\varphi\}$ of formulas: $\Gamma \vdash_{\mathbf{QCiore}} \varphi$ iff $(\forall)\Gamma \vdash_{\mathbf{QCiore}} (\forall)\varphi$, and $\Gamma \models_{\mathbf{QCiore}} \varphi$ iff $(\forall)\Gamma \models_{\mathbf{QCiore}} (\forall)\varphi$. From this, a general completeness result follows (see Corollary below).*

COROLLARY 7.22 (Completeness). *Let $\Gamma \cup \{\varphi\}$ be a set of formulas over Θ . Then $\Gamma \models_{\mathbf{QCiore}} \varphi$ implies $\Gamma \vdash_{\mathbf{QCiore}} \varphi$.*

Proof. It follows from Remark 7.21 and by Theorem 7.20. $\square$

THEOREM 7.23 (Compactness). *A theory T has a **QCiore**-model iff every finite subset of T has a **QCiore**-model.*

Proof. $(\Rightarrow)$: Immediate.

$(\Leftarrow)$: Since every subtheory of T has a **QCiore**-model, by hypothesis, then it follows that every finite subtheory of T is **QCiore**-consistent. Therefore T must be **QCiore**-consistent, by finitariness of **QCiore**, and so T has a **QCiore**-model, by Theorem 7.19. $\square$

COROLLARY 7.24. *Let $\Gamma \cup \{\varphi\}$ be a set of sentences. If $\Gamma \models_{\mathbf{QCiore}} \varphi$ then $\Gamma_0 \models_{\mathbf{QCiore}} \varphi$, for some finite $\Gamma_0 \subseteq \Gamma$.*

Proof. Suppose that for every finite $\Gamma_0 \subseteq \Gamma$ it is not the case of $\Gamma_0 \models_{\mathbf{QCiore}} \varphi$. By Proposition 7.15(i) we have that $T_0 = \Gamma_0 \cup \{\neg\varphi, \circ\varphi\}$ is **QCiore**-consistent and so, by Theorems 7.18 and 7.19, T_0 has a **QCiore**-model, for every $\Gamma_0 \subseteq \Gamma$ with Γ_0 finite. By Theorem 7.23, we have that $\Gamma \cup \{\neg\varphi, \circ\varphi\}$ has a **QCiore**-model. Therefore, it is not the case of $\Gamma \models_{\mathbf{QCiore}} \varphi$. $\square$

7.1. Some limitations of LFI2*. In this subsection, a brief comparison between **QCiore** and **LFI2***, the first-order version of **Ciore** proposed in Carnielli et al. (2000), will be made. It will be shown that the latter presents some limitations with respect to the former. As mentioned in the Introduction, it is important to observe that, unlike to what happens with **LFI2***, the quantifiers $\forall$ and $\exists$ are not interdefinable in **QCiore** in terms of the negation $\neg$. Indeed:

PROPOSITION 7.25. *The following schemas are not **QCiore**-valid:*

- (1) $\exists x \neg\varphi \rightarrow \neg\forall x\varphi$, (2) $\forall x \neg\varphi \rightarrow \neg\exists x\varphi$,
- (3) $\forall x\varphi \rightarrow \neg\exists x\neg\varphi$, (4) $\exists x\varphi \rightarrow \neg\forall x\neg\varphi$.

Proof. Let us consider the signature $\Theta = \langle P \rangle$ where P is an unary predicate symbol and consider the **QCiore**-structure $\mathfrak{A}$ where $A = \{a, b, c\}$ and $P^{\mathfrak{A}} = \langle a, \emptyset, \{b, c\} \rangle$. For any

variable x and any $e \in A$ let $[x \mapsto e] := \{s \in S(\mathfrak{A}) : s(x) = e\}$. Then, in $\mathfrak{A}$ we have

$$\begin{aligned} \|P(x)\|_{\mathfrak{A}}^{\mathfrak{A}} &= \langle [x \mapsto a], \emptyset, [x \mapsto b] \cup [x \mapsto c] \rangle \\ \|\forall x P(x)\|_{\oplus}^{\mathfrak{A}} &= \widehat{\exists x}([x \mapsto a]) = S(\mathfrak{A}) \\ \|\neg \forall x P(x)\|_{\ominus}^{\mathfrak{A}} &= \|\forall x P(x)\|_{\oplus}^{\mathfrak{A}} = S(\mathfrak{A}) \\ \|\neg P(x)\|_{\mathfrak{A}}^{\mathfrak{A}} &= \langle \emptyset, [x \mapsto a], [x \mapsto b] \cup [x \mapsto c] \rangle \\ \|\exists x \neg P(x)\|_{\ominus}^{\mathfrak{A}} &= \widehat{\forall x}([x \mapsto a]) = \emptyset. \end{aligned}$$

Therefore, it is not the case that $\mathfrak{A} \Vdash_{\mathbf{QCiore}} \exists x \neg \varphi \rightarrow \neg \forall x \varphi$, by Proposition 7.1 (iii). In a similar way, we can construct **QCiore**-structures in order to show that (2), (3), and (4) are not valid in **QCiore**. $\square$

This shows that **QCiore** differs from **LF12***, since the formulas $\neg \forall x \varphi$ and $\exists x \neg \varphi$ are equivalent in **LF12***, as well as the formulas $\neg \exists x \varphi$ and $\forall x \neg \varphi$. Given that universal quantification and existential quantification correspond to (infinitary) conjunctions and disjunctions, respectively, the latter equivalences in **LF12*** correspond to (infinitary) De Morgan rules. This is a somewhat counter-intuitive feature of **LF12***, given that **Ciore** does not satisfy the De Morgan rules. In Proposition 7.26 below it will be shown that this unexpected feature of **LF12*** produces, when combined with other natural features of **LF12***, undesirable and limitative theorems. Recall that **LF12** (and so **LF12***) has a primitive connective $\bullet$ for the inconsistency operator, such that $\bullet \alpha$ is equivalent to $\neg \circ \alpha$ as well as $\neg \bullet \alpha$ is equivalent to $\circ \alpha$. In the three-valued algebra, $\bullet(x) = 1$ if $x = \frac{1}{2}$, and 0 otherwise. This means that $\bullet \alpha$ is equivalent to $\alpha \wedge \neg \alpha$. Hence:

PROPOSITION 7.26. *Let $\varphi(x)$ be a formula in which x is the unique variable occurring free. The following formulas are theorems of **LF12***:*

- (1) $(\forall x \varphi(x) \wedge \exists x \neg \varphi(x)) \rightarrow \forall x \bullet \varphi(x)$;
- (2) $(\forall x \neg \varphi(x) \wedge \exists x \varphi(x)) \rightarrow \forall x \bullet \varphi(x)$.

Proof. (1) Consider the following derivation in **LF12***:

$$\begin{array}{l|l} 1. \forall x \varphi(x) & \text{hyp.} \\ 2. \exists x \neg \varphi(x) & \text{hyp.} \\ 3. \neg \forall x \varphi(x) & 2, \mathbf{LF12}^* \\ 4. \bullet \forall x \varphi(x) & 3, \mathbf{LF12}^* \\ 5. \forall x \bullet \varphi(x) & 4, \mathbf{LF12}^* \end{array}$$

Then, use the weak deduction metatheorem for **LF12***.

(2) It is proved analogously. $\square$

Observe that the key to derive $\neg \forall x \varphi(x)$ from the premisses in the derivation above was given in Step 3, by the interaction between the quantifiers allowed in **LF12***. On the other hand, Steps 4 and 5 correspond to expected properties of **LF12***.

The latter result shows that **LF12*** presents severe limitations in terms of being a first-order version of **Ciore**. The interaction between the (expected) axioms relating $\bullet$ with the quantifiers and the (unexpected) axioms stating the interdefinability of the quantifiers by means of $\neg$, imposes deep limitations to the models. Indeed, for any formula $\varphi(x)$ where x is the unique variable occurring free (in particular, if φ is any unary predicate symbol), let H be the set of truth-values corresponding to all the instances of $\varphi(x)$ assigned in a given

interpretation. Thus, if the set H is contained in $\{1, \frac{1}{2}\}$ or in $\{0, \frac{1}{2}\}$, then H must be $\{\frac{1}{2}\}$. That is, every instance of $\varphi(x)$ in such interpretation must get the value $\frac{1}{2}$. This seems to be an excessive restriction, which is not present in **QCiore**. These semantical limitations are reflected in Proposition 7.26, which shows that **LFI2*** derives some undesirable theorems. Because of these considerations, it can be argued that **QCiore** is a more faithful first-order version of **Ciore** than **LFI2*** is.

§8. QCiore-structures with equality. Given a first-order signature Θ , we shall denote $\Theta(\approx)$ the extension of Θ obtained by adding a binary predicate symbol $\approx$ that we call *equality*. As suggested by its name, $\approx$ will represent the identity predicate within the paraconsistent first-order logic **QCiore**.

DEFINITION 8.1. *The logic **QCiore**($\approx$) over the signature $\Theta(\approx)$ is the extension of **QCiore** over $\Theta(\approx)$ obtained by adding the following schemas (as usual, we write $(t \approx t')$ instead of $\approx(t, t')$):*

- $\forall x(x \approx x)$
- $\forall x \forall y((x \approx y) \rightarrow (\varphi \rightarrow \varphi[x \wr y]))$, if y is a variable free for x in φ

where $\varphi[x \wr y]$ denotes any formula obtained from φ by replacing some, but not necessarily all, free occurrences of x by y .

Let $\vec{x} = x_1, \dots, x_n$ be a list of n distinct variables, and let $\vec{t} = t_1, \dots, t_n$ be a list of terms. If φ is a formula depending at most on the variables $x_1, \dots, x_n$ then $\varphi[\vec{t}/\vec{x}]$ will denote the formula obtained from φ by simultaneously replacing the variable x_i by t_i , for $1 \leq i \leq n$.

PROPOSITION 8.2. ***QCiore**($\approx$) proves the following, for any terms t, t' and t'' :*

1. $\vdash (t \approx t)$,
2. $\vdash (t \approx t') \rightarrow (t' \approx t)$,
3. $\vdash (t \approx t') \rightarrow ((t' \approx t'') \rightarrow (t \approx t''))$,
4. $\vdash (\bigwedge_{i=1}^n (t_i \approx t'_i)) \rightarrow (f(t_1, \dots, t_n) \approx f(t'_1, \dots, t'_n))$ for every terms $t_1, t'_1, \dots, t_n, t'_n$ and any function symbol f of arity n ,
5. $\vdash (\bigwedge_{i=1}^n (t_i \approx t'_i)) \rightarrow (\varphi[\vec{t}/\vec{x}] \rightarrow \varphi[\vec{t}'/\vec{x}])$ for every terms $t_1, t'_1, \dots, t_n, t'_n$ and any formula without quantifiers φ depending at most on the variables $x_1, \dots, x_n$.

Proof. The proof is identical to the one for classical first-order logic with equality. For a proof of items 1-3 see, for instance, (Mendelson, 1987, Prop. 2.23). Items 4-5 are an easy consequence of axiom (**Ax12**) and the second axiom for $\approx$ presented in Definition 8.1. Indeed, let $\vec{y} = y_1, \dots, y_n$ be a list of n distinct variables such that $x_i \neq y_j$ for any i, j . Since φ has no quantifiers and it depends at most on the variables $x_1, \dots, x_n$, it follows that $\varphi(y_1/x_1)(y_2/x_2) \cdots (y_n/x_n) = \varphi[\vec{y}/\vec{x}]$. From this, and by applying n -times the second axiom in Definition 8.1 for $(x_1 \approx y_1), (x_2 \approx y_2), \dots, (x_n \approx y_n)$, together with axiom (**Ax12**) and the transitivity of $\rightarrow$, it follows that $\vdash (\bigwedge_{i=1}^n (x_i \approx y_i)) \rightarrow (\varphi \rightarrow \varphi[\vec{y}/\vec{x}])$. By Theorem 4.3, $\vdash \forall x_1 \forall y_1 \cdots \forall x_n \forall y_n ((\bigwedge_{i=1}^n (x_i \approx y_i)) \rightarrow (\varphi \rightarrow \varphi[\vec{y}/\vec{x}]))$. Finally the result follows by axiom (**Ax12**), using the fact that φ has no quantifiers and it depends at most on the variables $x_1, \dots, x_n$. $\square$

For any set A let $\Delta(A) := \{(a, b) \in A \times A : a = b\}$ be the *diagonal* of A . It is worth noting that a **QCiore**-structure $\mathfrak{A}$ over $\Theta(\approx)$ is a model of the sentence $\forall x(x \approx x)$, if and only if $\Delta(A) \subseteq \approx_{\oplus}^{\mathfrak{A}} \cup \approx_{\odot}^{\mathfrak{A}}$. This motivates the following:

DEFINITION 8.3. We say that a **QCiore**-structure $\mathfrak{A}$ over $\Theta(\approx)$ has a partial equality (or a pragmatic equality, or a paraconsistent equality) if

$$\approx_{\oplus}^{\mathfrak{A}} \cup \approx_{\odot}^{\mathfrak{A}} = \Delta(A).$$

In this case we say that $\mathfrak{A}$ is a **QCiore**($\approx$)-structure.

PROPOSITION 8.4. Let $\mathfrak{A}$ be a **QCiore**($\approx$)-structure. Then $\mathfrak{A}$ is a model of the logic **QCiore**($\approx$). That is, $\mathfrak{A}$ is a model of the axioms presented in Definition 8.1.

Proof. Straightforward, from the definition of **QCiore**($\approx$)-structure. $\square$

REMARK 8.5. Observe that a **QCiore**($\approx$)-structure $\mathfrak{A}$ validates a sentence $(t \approx t')$ iff the interpretations of t and t' in $\mathfrak{A}$ coincide. That is, the equality symbol has a standard or normal interpretation, according to the usual terminology adopted in first-order logic. However, a **QCiore**($\approx$)-structure $\mathfrak{A}$ is also allowed to validate a sentence of the form $\neg(t \approx t)$. This means that the equality symbol, even with a standard interpretation, is paraconsistent in the present approach. This is similar to the case for **QmbC** and other quantified **LFI**s with equality (see (Carnielli & Coniglio, 2016, chap. 7)).

DEFINITION 8.6. $\Gamma \cup \{\varphi\}$ be a set of formulas over $\Theta(\approx)$. We say that φ is a **QCiore**($\approx$)-consequence of Γ , denoted by $\Gamma \Vdash_{\mathbf{QCiore}(\approx)} \varphi$, if, for every **QCiore**($\approx$)-structure $\mathfrak{A}$, we have that: $\mathfrak{A} \Vdash_{\mathbf{QCiore}} \psi$ for every $\psi \in \Gamma$ implies that $\mathfrak{A} \Vdash_{\mathbf{QCiore}} \varphi$.

THEOREM 8.7. Let T be a maximal **QCiore**($\approx$)-consistent and **QCiore**($\approx$)-Henkin theory. Then T has a **QCiore**($\approx$)-model.

Proof. Let T be a theory which is **QCiore**($\approx$)-consistent and **QCiore**($\approx$)-Henkin. By Theorem 7.19 we have that T has a **QCiore**-model $\mathfrak{A}$ whose domain is the set A of all the closed terms over the signature $\Theta(\approx)$. Moreover, $t^{\mathfrak{A}}(s) = t$ for every closed term t and every $s \in S(\mathfrak{A})$, and

$$\begin{array}{lll} (1) & \|\varphi\|^{\mathfrak{A}} = 1 & \text{iff} & \varphi \wedge \circ\varphi \in T \\ (2) & \|\varphi\|^{\mathfrak{A}} = 0 & \text{iff} & \neg\varphi \wedge \circ\varphi \in T \\ (3) & \|\varphi\|^{\mathfrak{A}} = \frac{1}{2} & \text{iff} & \varphi \wedge \neg\varphi \in T \end{array}$$

for every sentence φ in $\text{Sent}(\Theta(\approx))$. Let $\equiv$ be the following relation on A : $t \equiv t'$ iff $(t \approx t') \in T$. By Proposition 8.2(1)-(3) it follows that $\equiv$ is an equivalence relation. Let $\bar{a}$ be the equivalence class of $a \in A$ determined by $\equiv$. Consider a new **QCiore**-structure $\mathfrak{A}^*$ defined as follows:

- the domain of $\mathfrak{A}^*$ is the set $A^* = A/\equiv$ of equivalence classes;
- if R is a predicate symbol of arity n then $R^{\mathfrak{A}^*} = \langle R_{\oplus}^{\mathfrak{A}}, R_{\odot}^{\mathfrak{A}}, R_{\circ}^{\mathfrak{A}} \rangle$ is defined as follows:

$$\begin{array}{lll} (\bar{a}_1, \dots, \bar{a}_n) \in R_{\oplus}^{\mathfrak{A}^*} & \text{iff} & R(a_1, \dots, a_n) \wedge \circ R(a_1, \dots, a_n) \in T \\ (\bar{a}_1, \dots, \bar{a}_n) \in R_{\odot}^{\mathfrak{A}^*} & \text{iff} & \neg R(a_1, \dots, a_n) \wedge \circ R(a_1, \dots, a_n) \in T \\ (\bar{a}_1, \dots, \bar{a}_n) \in R_{\circ}^{\mathfrak{A}^*} & \text{iff} & R(a_1, \dots, a_n) \wedge \neg R(a_1, \dots, a_n) \in T \end{array}$$

- if f is a function symbol of arity n then $f^{\mathfrak{A}^*}(\bar{a}_1, \dots, \bar{a}_n) = \overline{f(a_1, \dots, a_n)}$;
- if c is a constant symbol then $c^{\mathfrak{A}^*} = \bar{c}$.

Fact: The **QCiore**-structure $\mathfrak{A}^*$ is well-defined.

Indeed, suppose that R is a predicate symbol of arity n and $a_i \equiv a'_i$ for $1 \leq i \leq n$ (recalling that, by definition, every a_i and a'_i is a closed term). Then, by Proposition 8.2(5), $R(a_1, \dots, a_n) \wedge \circ R(a_1, \dots, a_n) \in T$ iff $R(a'_1, \dots, a'_n) \wedge \circ R(a'_1, \dots, a'_n) \in T$. Hence $R_{\oplus}^{\mathfrak{A}}$ is well-defined. Analogously it can be proven that $R_{\ominus}^{\mathfrak{A}}$ and $R_{\odot}^{\mathfrak{A}}$ are well-defined. Now, if f is a function symbol of arity n and $a_i \equiv a'_i$ for $1 \leq i \leq n$ then, by Proposition 8.2(4), $(f(a_1, \dots, a_n) \approx f(a'_1, \dots, a'_n)) \in T$, therefore $f^{\mathfrak{A}^*} : (A^*)^n \rightarrow A^*$ is a well-defined function. This proves the **Fact**.

Observe that $\approx^{\mathfrak{A}^*}$ is a paraconsistent equality. Indeed, $(\bar{a}, \bar{a}') \in \approx_{\oplus}^{\mathfrak{A}^*} \cup \approx_{\odot}^{\mathfrak{A}^*}$ iff, by definition of the interpretation of the predicates in $\mathfrak{A}^*$ and $\mathfrak{A}$, $(a, a') \in \approx_{\oplus}^{\mathfrak{A}} \cup \approx_{\odot}^{\mathfrak{A}}$ iff $(a \approx a') \in T$ iff $\bar{a} = \bar{a}'$. Thus, $\mathfrak{A}^*$ is a **QCiore**($\approx$)-structure. Finally, $\mathfrak{A}^*$ is a **QCiore**($\approx$)-model for T . Indeed, let $s \in S(\mathfrak{A})$ and $s^* \in S(\mathfrak{A}^*)$ be defined by $s^*(x) = \bar{s}(x)$. Then, it can be proved by induction on the complexity of the formula φ that

$$\mathfrak{A} \Vdash \varphi[s] \text{ iff } \mathfrak{A}^* \Vdash \varphi[s^*].$$

Then, $\mathfrak{A}^* \Vdash_{\mathbf{QCiore}} T$. That is, $\mathfrak{A}^*$ is a **QCiore**($\approx$)-model of T . □

THEOREM 8.8 (Soundness and completeness of **QCiore**($\approx$)). *Let $\Gamma \cup \{\varphi\}$ be a set of sentences over $\Theta(\approx)$. Then $\Gamma \vdash_{\mathbf{QCiore}(\approx)} \varphi$ if and only if $\Gamma \Vdash_{\mathbf{QCiore}(\approx)} \varphi$.*

Proof. ($\Rightarrow$): It is consequence of Theorem 7.12 and Proposition 8.4.

($\Leftarrow$): Suppose that φ is not derivable from Γ in **QCiore**($\approx$). Then, by Proposition 7.15, we have that the theory

$$T' = \Gamma \cup \{\neg\varphi, \circ\varphi\} \cup Ax(\approx)$$

(where $Ax(\approx)$ is the set of axiom schemas for equality given in Definition 8.1) is **QCiore**-consistent. By Theorem 7.18, there is a theory T over a signature $\Theta(\approx)_C$ (obtained from $\Theta(\approx)$ by adding a set C of new constants) which is maximal **QCiore**-consistent and **QCiore**-Henkin, and such that $T' \subseteq T$. Therefore, T is a maximal **QCiore**($\approx$)-consistent and **QCiore**($\approx$)-Henkin theory. Thus, T has a **QCiore**($\approx$)-model $\mathfrak{A}$ over $\Theta(\approx)_C$, by Theorem 8.7. Let $\mathfrak{A}'$ be the reduct of $\mathfrak{A}$ to $\Theta(\approx)$. Then, $\mathfrak{A}'$ is a **QCiore**($\approx$)-structure over $\Theta(\approx)_C$ such that $\mathfrak{A}' \Vdash_{\mathbf{QCiore}} \Gamma$ and $\mathfrak{A}' \Vdash_{\mathbf{QCiore}} \neg\varphi \wedge \circ\varphi$, and so $\mathfrak{A}' \not\Vdash_{\mathbf{QCiore}} \varphi$. Therefore, we have that $\Gamma \not\Vdash_{\mathbf{QCiore}(\approx)} \varphi$. □

As it was done for **QCiore**, Theorem 8.8 can be easily extended to formulas (possibly) having free variables. It is not difficult to obtain a Compactness Theorem for **QCiore**($\approx$).

§9. Some model-theoretic results for QCiore. In what follows, we shall assume that in every signature we have at least the binary predicate symbol $\approx$ of equality. Moreover, every **QCiore**-structure will be assumed to be a **QCiore**($\approx$) structure, in the sense of Definition 8.3. Thus, we will simply write Θ instead of $\Theta(\approx)$ to refer to a given first-order signature. The notion of partial equality together with the Compactness Theorem allow us to make model-theoretic constructions analogous to the ones made for Tarskian structures. To begin with, the following theorems can be easily obtained (the details are left to the reader).

THEOREM 9.1. *If a theory T has arbitrarily large finite **QCiore**($\approx$)-models, then it has an infinite **QCiore**($\approx$)-model.*

THEOREM 9.2. *If a theory T over a signature Θ has infinite $\mathbf{QCiore}(\approx)$ -models, then it has infinite $\mathbf{QCiore}(\approx)$ -models of any given cardinality $\geq \aleph_0 + |\mathcal{L}(\Theta)|$, where $|\mathcal{L}(\Theta)|$ denotes the cardinality of the first-order language $\mathcal{L}(\Theta)$ generated by Θ .*

Let $\mathfrak{A}$ and $\mathfrak{B}$ be two $\mathbf{QCiore}$ -structures over Θ . We say that $\mathfrak{A}$ and $\mathfrak{B}$ are *elementarily equivalent*, denoted by $\mathfrak{A} \equiv \mathfrak{B}$, if for every sentence φ , if $\mathfrak{A} \Vdash_{\mathbf{QCiore}} \varphi$ then $\mathfrak{B} \Vdash_{\mathbf{QCiore}} \varphi$.

PROPOSITION 9.3. *Assume that $\mathfrak{A} \equiv \mathfrak{B}$. Then, for every sentence φ it holds: $\mathfrak{A} \Vdash_{\mathbf{QCiore}} \varphi$ iff $\mathfrak{B} \Vdash_{\mathbf{QCiore}} \varphi$.*

Proof. The ‘only if’ part is immediate, by definition of $\equiv$. Now, suppose that φ is a sentence such that $\mathfrak{A} \not\Vdash_{\mathbf{QCiore}} \varphi$. By Corollary 7.8 it follows that $\mathfrak{A} \Vdash_{\mathbf{QCiore}} \neg\varphi \wedge \circ\varphi$. By hypothesis, $\mathfrak{B} \Vdash_{\mathbf{QCiore}} \neg\varphi \wedge \circ\varphi$ and so, using again Corollary 7.8, it follows that $\mathfrak{B} \not\Vdash_{\mathbf{QCiore}} \varphi$. $\square$

$\mathfrak{A}$ is said to be a *sub- $\mathbf{QCiore}$ -structure* of $\mathfrak{B}$, denoted by $\mathfrak{A} \subseteq \mathfrak{B}$, if it is verified: (1) $c^{\mathfrak{A}} = c^{\mathfrak{B}}$ for every constant symbol c , (2) $f^{\mathfrak{A}} = f^{\mathfrak{B}}|_{A^n}$ for every n -ary function symbol f ; and (3) $R_{\#}^{\mathfrak{A}} = R_{\#}^{\mathfrak{B}} \cap A^n$ for every $\# \in \{\oplus, \odot, \ominus\}$ and for every n -ary predicate symbol R .

REMARK 9.4. *Suppose that $\mathfrak{A} \subseteq \mathfrak{B}$. If $s \in S(\mathfrak{A})$ then it induces a unique $\bar{s} \in S(\mathfrak{B})$ given by $\bar{s}(x) = s(x)$ for every $x \in \mathcal{V}$. The subset $\{\bar{s} : s \in S(\mathfrak{A})\}$ of $S(\mathfrak{B})$ will be denoted by $\bar{S}(\mathfrak{A})$. Note that $\bar{s}_x^a = \bar{s}_x^a$ for every $s \in S(\mathfrak{A})$, $x \in \mathcal{V}$ and $a \in A$. For every formula φ and $\# \in \{\oplus, \odot, \ominus\}$ let*

$$\overline{\|\varphi\|_{\#}^{\mathfrak{A}}} = \{\bar{s} : s \in \|\varphi\|_{\#}^{\mathfrak{A}}\}.$$

DEFINITION 9.5. *We say that $\mathfrak{A}$ is a elementary sub- $\mathbf{QCiore}$ -structure of $\mathfrak{B}$, denoted by $\mathfrak{A} \prec \mathfrak{B}$, if $\mathfrak{A} \subseteq \mathfrak{B}$ and for every formula φ it holds:*

$$\overline{\|\varphi\|_{\#}^{\mathfrak{A}}} = \|\varphi\|_{\#}^{\mathfrak{B}} \cap \bar{S}(\mathfrak{A})$$

where $\# \in \{\oplus, \odot, \ominus\}$. Equivalently, $\mathfrak{A} \prec \mathfrak{B}$ iff for every formula φ and every $s \in S(\mathfrak{A})$ we have that

$$\mathfrak{A} \Vdash_{\mathbf{QCiore}} \varphi[s] \text{ iff } \mathfrak{B} \Vdash_{\mathbf{QCiore}} \varphi[\bar{s}]$$

From the definition, it clearly follows that $\mathfrak{A} \prec \mathfrak{B}$ implies $\mathfrak{A} \equiv \mathfrak{B}$.

LEMMA 9.6 (Tarski-like conditions). *Let $\mathfrak{A}$ and $\mathfrak{B}$ be two $\mathbf{QCiore}$ -structures such that $\mathfrak{A} \subseteq \mathfrak{B}$. Then, the following conditions are equivalent:*

(i) $\mathfrak{A} \prec \mathfrak{B}$,

(ii) for all $s \in S(\mathfrak{A})$ the following conditions hold:

(TC1) *If $\bar{s} \in \|\exists x\varphi\|_{\oplus}^{\mathfrak{B}}$, then there exist $a, b \in A$ such that $\bar{s}_x^a \notin \|\varphi\|_{\odot}^{\mathfrak{B}}$ and $\bar{s}_x^b \notin \|\varphi\|_{\odot}^{\mathfrak{B}}$*

(TC2) *If $\bar{s} \notin \|\exists x\varphi\|_{\odot}^{\mathfrak{B}}$, then there exists $a \in A$ such that $\bar{s}_x^a \notin \|\varphi\|_{\odot}^{\mathfrak{B}}$*

(TC3) *If $\bar{s} \in \|\forall x\varphi\|_{\oplus}^{\mathfrak{B}}$, then there exists $a \in A$ such that $\bar{s}_x^a \in \|\varphi\|_{\oplus}^{\mathfrak{B}}$*

(TC4) *If $\bar{s} \in \|\forall x\varphi\|_{\odot}^{\mathfrak{B}}$, then there exists $a \in A$ such that $\bar{s}_x^a \in \|\varphi\|_{\odot}^{\mathfrak{B}}$.*

Proof. (ii) $\Rightarrow$ (i): By induction on the complexity of the formula φ it will be proven that

$$\overline{\|\varphi\|_{\#}^{\mathfrak{A}}} = \|\varphi\|_{\#}^{\mathfrak{B}} \cap \bar{S}(\mathfrak{A}).$$

Base step: φ is an atomic formula $P(t_1, \dots, t_n)$. Then, for each $\# \in \{\oplus, \odot, \ominus\}$: $s \in \|\varphi\|_{\#}^{\mathfrak{A}}$ iff $(t_1^{\mathfrak{A}}[s], \dots, t_n^{\mathfrak{A}}[s]) \in P_{\#}^{\mathfrak{A}}$ iff $(t_1^{\mathfrak{B}}[\bar{s}], \dots, t_n^{\mathfrak{B}}[\bar{s}]) \in P_{\#}^{\mathfrak{B}}$ (since $\mathfrak{A} \subseteq \mathfrak{B}$) iff $\bar{s} \in \|\varphi\|_{\#}^{\mathfrak{B}}$.

Inductive step: Assume that $\overline{\|\psi\|_{\#}^{\mathfrak{A}}} = \|\psi\|_{\#}^{\mathfrak{B}} \cap \bar{S}(\mathfrak{A})$ for every ψ with complexity $\leq k$. Let φ with complexity $k+1$, and let $s \in S(\mathfrak{A})$.

(1) If φ is one of $\neg\psi$, $\circ\psi$, $\psi \wedge \chi$, $\psi \vee \chi$ or $\psi \rightarrow \chi$, it is proved easily by using the induction hypothesis.

(2) If φ is $\exists x\psi$ then:

(2.1) $s \in \|\exists x\psi\|_{\oplus}^{\mathfrak{A}}$ only if $s \notin \widehat{\nabla x}(\|\psi\|_{\oplus}^{\mathfrak{A}}) \cup \widehat{\nabla x}(\|\psi\|_{\odot}^{\mathfrak{A}})$, by Proposition 6.6, only if there exist $a, b \in A$ such that $s_x^a \notin \|\psi\|_{\oplus}^{\mathfrak{A}}$ and $s_x^b \notin \|\psi\|_{\odot}^{\mathfrak{A}}$, by Definition 6.4, only if there exist $a, b \in A$ such that $\bar{s}_x^a \notin \|\psi\|_{\oplus}^{\mathfrak{B}}$ and $\bar{s}_x^b \notin \|\psi\|_{\oplus}^{\mathfrak{B}}$ (by I.H. and since $s_x^a, s_x^b \in S(\mathfrak{A})$), only if there exist $a, b \in B$ such that $\bar{s}_x^a \notin \|\psi\|_{\oplus}^{\mathfrak{B}}$ and $\bar{s}_x^b \notin \|\psi\|_{\oplus}^{\mathfrak{B}}$ (since $A \subseteq B$), only if $\bar{s} \in \|\exists x\psi\|_{\oplus}^{\mathfrak{B}}$, by Proposition 6.6. Conversely: $\bar{s} \in \|\exists x\psi\|_{\oplus}^{\mathfrak{B}}$ only if there exist $a, b \in A$ such that $\bar{s}_x^a \notin \|\psi\|_{\oplus}^{\mathfrak{B}}$ and $\bar{s}_x^b \notin \|\psi\|_{\oplus}^{\mathfrak{B}}$ (by (TC1)) only if $s_x^a \notin \|\psi\|_{\oplus}^{\mathfrak{A}}$ and $s_x^b \notin \|\psi\|_{\odot}^{\mathfrak{A}}$ (by I.H. and since $s_x^a, s_x^b \in S(\mathfrak{A})$) only if $s \in \|\exists x\psi\|_{\oplus}^{\mathfrak{A}}$.

(2.2) Assume that $s \in \|\exists x\psi\|_{\odot}^{\mathfrak{A}}$, and suppose that $\bar{s} \notin \|\exists x\psi\|_{\oplus}^{\mathfrak{B}}$. By (TC2), there exists $a \in A$ such that $\bar{s}_x^a \notin \|\psi\|_{\oplus}^{\mathfrak{B}}$. By I.H., $s_x^a \notin \|\psi\|_{\odot}^{\mathfrak{A}}$ for some $a \in A$, a contradiction. Then $\|\exists x\psi\|_{\odot}^{\mathfrak{A}} \subseteq \|\exists x\psi\|_{\oplus}^{\mathfrak{B}} \cap \bar{S}(\mathfrak{A})$. Conversely: $\bar{s} \in \|\exists x\psi\|_{\oplus}^{\mathfrak{B}} = \widehat{\nabla x}(\|\psi\|_{\oplus}^{\mathfrak{B}})$ only if $\bar{s}_x^a \in \|\psi\|_{\oplus}^{\mathfrak{B}}$ for every $a \in B$ only if $\bar{s}_x^a \in \|\psi\|_{\oplus}^{\mathfrak{B}}$ for every $a \in A$ (since $A \subseteq B$) only if $s_x^a \in \|\psi\|_{\odot}^{\mathfrak{A}}$ for every $a \in A$ (by I.H.) only if $s \in \widehat{\nabla x}(\|\psi\|_{\odot}^{\mathfrak{A}}) = \|\exists x\psi\|_{\odot}^{\mathfrak{A}}$.

(2.3) $s \in \|\exists x\psi\|_{\oplus}^{\mathfrak{A}}$ iff $s \notin \|\exists x\psi\|_{\oplus}^{\mathfrak{A}} \cup \|\exists x\psi\|_{\odot}^{\mathfrak{A}}$ iff $\bar{s} \notin \|\exists x\psi\|_{\oplus}^{\mathfrak{B}} \cup \|\exists x\psi\|_{\odot}^{\mathfrak{B}}$ (by (2.1) and (2.2)) iff $\bar{s} \in \|\exists x\psi\|_{\oplus}^{\mathfrak{B}}$.

(3) If φ is $\forall x\psi$ then:

(3.1) $s \in \|\forall x\psi\|_{\oplus}^{\mathfrak{A}} = \widehat{\nabla x}(\|\psi\|_{\oplus}^{\mathfrak{A}}) - \widehat{\nabla x}(\|\psi\|_{\odot}^{\mathfrak{A}})$, only if: (1) $s \in \widehat{\nabla x}(\|\psi\|_{\oplus}^{\mathfrak{A}})$ and (2) $s \notin \widehat{\nabla x}(\|\psi\|_{\odot}^{\mathfrak{A}})$, only if $s_x^a \notin \|\psi\|_{\odot}^{\mathfrak{A}}$, for every $a \in A$ (by (2)), only if $\bar{s}_x^a \notin \|\psi\|_{\oplus}^{\mathfrak{B}}$, for every $a \in A$ (by I.H.), only if $\bar{s} \notin \|\forall x\psi\|_{\oplus}^{\mathfrak{B}} = \widehat{\nabla x}(\|\psi\|_{\oplus}^{\mathfrak{B}})$ (by (TC4)), only if $\bar{s} \in \widehat{\nabla x}(\|\psi\|_{\oplus}^{\mathfrak{B}})$ (by (1) and I.H.) only if $\bar{s} \in \|\forall x\psi\|_{\oplus}^{\mathfrak{B}} = \widehat{\nabla x}(\|\psi\|_{\oplus}^{\mathfrak{B}}) - \widehat{\nabla x}(\|\psi\|_{\odot}^{\mathfrak{B}})$. Conversely: $\bar{s} \in \|\forall x\psi\|_{\oplus}^{\mathfrak{B}} = \widehat{\nabla x}(\|\psi\|_{\oplus}^{\mathfrak{B}}) - \widehat{\nabla x}(\|\psi\|_{\odot}^{\mathfrak{B}})$ only if: (1) $\bar{s} \in \widehat{\nabla x}(\|\psi\|_{\oplus}^{\mathfrak{B}})$ and (2) $\bar{s} \notin \widehat{\nabla x}(\|\psi\|_{\odot}^{\mathfrak{B}})$, only if there exists $a \in A$ such that $s_x^a \in \|\psi\|_{\oplus}^{\mathfrak{A}}$ (by (TC3) and I.H.), only if $s \in \widehat{\nabla x}(\|\psi\|_{\oplus}^{\mathfrak{A}})$, only if $s \notin \widehat{\nabla x}(\|\psi\|_{\odot}^{\mathfrak{A}})$ (by (2), $A \subseteq B$ and I.H.), only if $s \in \|\forall x\psi\|_{\oplus}^{\mathfrak{A}} = \widehat{\nabla x}(\|\psi\|_{\oplus}^{\mathfrak{A}}) - \widehat{\nabla x}(\|\psi\|_{\odot}^{\mathfrak{A}})$.

(3.2) Suppose that $\bar{s} \notin \|\forall x\psi\|_{\oplus}^{\mathfrak{B}}$. Since $\|\forall x\psi\|_{\oplus}^{\mathfrak{B}} = \|\exists x\psi\|_{\oplus}^{\mathfrak{B}}$ then, by (TC2), there exists $a \in A$ such that $\bar{s}_x^a \notin \|\psi\|_{\oplus}^{\mathfrak{B}}$. By I.H., $s_x^a \notin \|\psi\|_{\odot}^{\mathfrak{A}}$ for some $a \in A$, hence $s \notin \widehat{\nabla x}(\|\psi\|_{\odot}^{\mathfrak{A}}) = \|\forall x\psi\|_{\odot}^{\mathfrak{A}}$. Then $\|\forall x\psi\|_{\odot}^{\mathfrak{A}} \subseteq \|\forall x\psi\|_{\oplus}^{\mathfrak{B}} \cap \bar{S}(\mathfrak{A})$. Conversely, $\bar{s} \in \|\forall x\psi\|_{\oplus}^{\mathfrak{B}}$ only if $\bar{s} \in \widehat{\nabla x}(\|\psi\|_{\oplus}^{\mathfrak{B}})$, only if $\bar{s}_x^a \in \|\psi\|_{\oplus}^{\mathfrak{B}}$ for every $a \in B$, only if $\bar{s}_x^a \in \|\psi\|_{\oplus}^{\mathfrak{B}}$ for every $a \in A$ (since $A \subseteq B$ and by I.H.), only if $s \in \widehat{\nabla x}(\|\psi\|_{\odot}^{\mathfrak{A}}) = \|\forall x\psi\|_{\odot}^{\mathfrak{A}}$.

(3.3) The proof for $\|\forall x\psi\|_{\odot}^{\mathfrak{A}}$ is similar to the case (2.3). □

Let α be an ordinal number and let Θ a first-order signature. A *chain of **QCiore**-structures of length α* is an increasing sequence of **QCiore**-structures $\mathfrak{A}_0 \subseteq \mathfrak{A}_1 \subseteq \dots \subseteq \mathfrak{A}_\beta \subseteq \dots$ (for $\beta < \alpha$) over Θ . The *union* of a chain as above is the **QCiore**-structure $\mathfrak{A} = \bigcup_{\beta < \alpha} \mathfrak{A}_\beta$ over Θ such that: (1) the domain of $\mathfrak{A}$ is the union $A = \bigcup_{\beta < \alpha} A_\beta$ of the respective domains of the **QCiore**-structures $\mathfrak{A}_\beta$; (2) $R_{\#}^{\mathfrak{A}} := \bigcup_{\beta < \alpha} R_{\#}^{\mathfrak{A}_\beta}$ for $\# \in \{\oplus, \odot, \ominus\}$ and for every n -ary predicate symbol $R \in \mathcal{P}_n$; (3) $f^{\mathfrak{A}} := \bigcup_{\beta < \alpha} f^{\mathfrak{A}_\beta}$ for every n -ary function symbol f ; and (4) $c^{\mathfrak{A}} := c^{\mathfrak{A}_0}$ for every constant symbol c . Note that $f^{\mathfrak{A}}$ is a function, since the chain

is increasing. Moreover, if $R \in \mathcal{P}_n$ then $R^{\mathfrak{A}} = \langle R_{\oplus}^{\mathfrak{A}}, R_{\ominus}^{\mathfrak{A}}, R_{\odot}^{\mathfrak{A}} \rangle$ is a triple over A^n . Then $\mathfrak{A}$ is, in fact, a **QCiore**-structure over Θ .

REMARK 9.7. Recall that $\mathfrak{A}$ is a **QCiore**($\approx$)-structure if $\approx_{\oplus}^{\mathfrak{A}} \cup \approx_{\odot}^{\mathfrak{A}} = \{(a, b) \in A \times A : a = b\}$. Then, it is easy to see that the union of a chain of **QCiore**($\approx$)-structures is a **QCiore**($\approx$)-structure.

Using union of chains, it can be proved a version for **QCiore**-structures of a well-known theorem obtained in 1957 by Tarski and Vaught, which generalizes the downward Löwenheim-Skolem-Tarski theorem:

THEOREM 9.8 (Tarski and Vaught theorem for **QCiore**-structures). *Let Θ be a first-order signature. Let $\mathfrak{B}$ be a **QCiore**-structure over Θ of cardinality α , and let β be a cardinal such that $|\mathcal{L}(\Theta)| \leq \beta \leq \alpha$. Then, there exists a **QCiore**-structure $\mathfrak{A}$ over Θ of cardinality β such that $\mathfrak{A} \prec \mathfrak{B}$.*

Proof. Let $X \subseteq B$ of cardinality β . We shall construct inductively a denumerable sequence of sets $A_0 \subseteq A_1 \subseteq \dots \subseteq A_n \subseteq \dots$ as follows: $A_0 = X \cup \{c^{\mathfrak{B}} : c \in \mathcal{C}\}$. Now, suppose that A_n was already defined. Then:

(LS0) if $f \in \mathcal{F}_k$ then $f^{\mathfrak{B}}(a_1, \dots, a_k) \in A_{n+1}$ for all $(a_1, \dots, a_k) \in A_n^k$.

Moreover, for every formula φ and every $s \in S(\mathfrak{B})$ taking values in A_n :

(LS1) if $s \in \|\exists x \varphi\|_{\oplus}^{\mathfrak{B}}$, we choose $a, b \in B$ such that $s_x^a \notin \|\varphi\|_{\ominus}^{\mathfrak{B}}$ and $s_x^b \notin \|\varphi\|_{\odot}^{\mathfrak{B}}$,

(LS2) if $s \notin \|\exists x \varphi\|_{\odot}^{\mathfrak{B}}$, we choose $c \in B$ such that $s_x^c \notin \|\varphi\|_{\oplus}^{\mathfrak{B}}$,

(LS3) if $s \in \|\forall x \varphi\|_{\oplus}^{\mathfrak{B}}$, we choose $d \in B$ such that $s_x^d \in \|\varphi\|_{\oplus}^{\mathfrak{B}}$,

(LS4) if $s \in \|\forall x \varphi\|_{\odot}^{\mathfrak{B}}$, we choose $e \in B$ such that $s_x^e \in \|\varphi\|_{\odot}^{\mathfrak{B}}$.

Note that the existence of the elements a, b, c, d, e is guaranteed by Proposition 6.6. Besides, we add to the set A_{n+1} the elements $\{a, b\}$ (or the elements c, d or e) as appropriate. Let $\mathfrak{A}$ be the **QCiore**-structure with domain $A = \bigcup_{n \geq 0} A_n$ such that

- $c^{\mathfrak{A}} := c^{\mathfrak{B}}$, for every $c \in \mathcal{C}$,
- if $f \in \mathcal{F}_k$, let $f^{\mathfrak{A}} := f^{\mathfrak{B}}|_{A^k}$ (which is well-defined, by (LS0))
- if $R \in \mathcal{P}_k$, let $R_{\#}^{\mathfrak{A}} := R_{\#}^{\mathfrak{B}} \cap A^k$ for every $\# \in \{\oplus, \odot, \ominus\}$.

Claim 1. The domain A has cardinality β and $\mathfrak{A} \subseteq \mathfrak{B}$.

It is immediate from the definitions.

Claim 2. Given $s \in S(\mathfrak{A})$, there exists $s' : \mathcal{V} \rightarrow A_k$ (for some $k \in \mathbb{N}$) such that $\|\varphi\|_{\oplus}^{\mathfrak{B}}(s) = \|\varphi\|_{\oplus}^{\mathfrak{B}}(s')$.

It follows by the fact that every formula has a finite number of free variables, and by Proposition 7.5.

Claim 3. $\mathfrak{A}$ and $\mathfrak{B}$ satisfy Conditions (TC1)–(TC4) of Lemma 9.6 (Tarski-like conditions). In order to prove this, let $s \in S(\mathfrak{A})$.

Condition (TC1): If $\bar{s} \in \|\exists x \varphi\|_{\oplus}^{\mathfrak{B}}$ then, by **Claim 2** and (LS1), there are $a, b \in A_{k+1} \subseteq A$ such that $\bar{s}_x^a \notin \|\varphi\|_{\ominus}^{\mathfrak{B}}$ and $\bar{s}_x^b \notin \|\varphi\|_{\odot}^{\mathfrak{B}}$.

Condition (TC2): If $\bar{s} \notin \|\varphi\|_{\odot}^{\mathfrak{B}}$, by **Claim 2** and (LS2), there exists $a \in A_{k+1} \subseteq A$ such that $\bar{s}_x^a \notin \|\varphi\|_{\odot}^{\mathfrak{B}}$.

Condition (TC3): If $\bar{s} \in \|\forall x\varphi\|_{\oplus}^{\mathfrak{B}}$, by **Claim 2** and (LS3), there exists $a \in A_{k+1} \subseteq A$ such that $\bar{s}_x^a \in \|\varphi\|_{\oplus}^{\mathfrak{B}}$.

Condition (TC4): If $\bar{s} \in \|\forall x\varphi\|_{\ominus}^{\mathfrak{B}}$, by **Claim 2** and (LS4), there exists $a \in A_{k+1} \subseteq A$ such that $\bar{s}_x^a \in \|\varphi\|_{\ominus}^{\mathfrak{B}}$.

As a consequence of Lemma 9.6, $\mathfrak{A} \prec \mathfrak{B}$. □

An *elementary chain* is a chain of **QCiore**-structures such that $\mathfrak{A}_k \prec \mathfrak{A}_n$ whenever $k < n$. The Elementary Chain Theorem can be adapted to **QCiore**-structures:

THEOREM 9.9 (Elementary Chain Theorem). *Let $\langle \mathfrak{A}_n : n \in \mathbb{N} \rangle$ be an elementary chain of **QCiore**-structures. Then $\mathfrak{A}_k \prec \bigcup_n \mathfrak{A}_n$ for all $k \in \mathbb{N}$.*

Proof. Let us prove that $\overline{\|\varphi\|_{\#}^{\mathfrak{A}_k}} = \|\varphi\|_{\#}^{\mathfrak{A}} \cap \bar{S}(\mathfrak{A}_k)$ for every $k \in \mathbb{N}$ and $\# \in \{\oplus, \odot, \ominus\}$ by induction on the complexity of the formula φ . Thus, let $k \in \mathbb{N}$ and $s \in S(\mathfrak{A}_k)$.

Basic step: Let φ be the atomic formula $P(t_1, \dots, t_n)$, and $\# \in \{\oplus, \odot, \ominus\}$. Clearly $t^{\mathfrak{A}}[\bar{s}] = t^{\mathfrak{A}_k}[s]$ for each term t . Hence

$$\begin{aligned} \|P(t_1, \dots, t_n)\|_{\#}^{\mathfrak{A}}(\bar{s}) &= P^{\mathfrak{A}}(t_1^{\mathfrak{A}}[\bar{s}], \dots, t_n^{\mathfrak{A}}[\bar{s}]) \\ &= P^{\mathfrak{A}_k}(t_1^{\mathfrak{A}_k}[s], \dots, t_n^{\mathfrak{A}_k}[s]) \\ &= \|P(t_1, \dots, t_n)\|_{\#}^{\mathfrak{A}_k}(s) \end{aligned}$$

Inductive step: Assume that $\overline{\|\varphi\|_{\#}^{\mathfrak{A}_k}} = \|\varphi\|_{\#}^{\mathfrak{A}} \cap \bar{S}(\mathfrak{A}_k)$ for every $k \in \mathbb{N}$, $\# \in \{\oplus, \odot, \ominus\}$ and ψ with complexity $\leq n$. Let φ with complexity $n+1$, let $k \in \mathbb{N}$ and $s \in S(\mathfrak{A}_k)$.

(1) If φ is one of $\neg\psi$, $\odot\psi$, $\psi \wedge \chi$, $\psi \vee \zeta$ or $\psi \rightarrow \zeta$, it is proved easily by using the induction hypothesis.

(2) If φ is $\exists x\psi$, then:

(2.1) If $\bar{s} \in \|\exists x\psi\|_{\oplus}^{\mathfrak{A}}$ then there exist $a, b \in A = \bigcup_n A_n$ such that $\bar{s}_x^a \notin \|\psi\|_{\odot}^{\mathfrak{A}}$ and $\bar{s}_x^b \notin \|\psi\|_{\odot}^{\mathfrak{A}}$. Thus, there exist $j \in \mathbb{N}$ and $a, b \in A_j$ such that $\bar{s}_x^a \notin \|\psi\|_{\odot}^{\mathfrak{A}_j}$ and $\bar{s}_x^b \notin \|\psi\|_{\odot}^{\mathfrak{A}_j}$. If $j \leq k$ then there exist $a, b \in A_k$ such that $\bar{s}_x^a \notin \|\psi\|_{\odot}^{\mathfrak{A}_k}$ and $\bar{s}_x^b \notin \|\psi\|_{\odot}^{\mathfrak{A}_k}$, by I.H., then $s \in \|\exists x\psi\|_{\oplus}^{\mathfrak{A}_k}$. Otherwise, if $k < j$ then $\mathfrak{A}_k \prec \mathfrak{A}_j$. Note that $s, s_x^a, s_x^b \in S(\mathfrak{A}_j)$ up to names, in the sense of Remark 9.4. By I.H., $s_x^a \notin \|\psi\|_{\odot}^{\mathfrak{A}_j}$ and $s_x^b \notin \|\psi\|_{\odot}^{\mathfrak{A}_j}$, then $s \in \|\exists x\psi\|_{\oplus}^{\mathfrak{A}_j}$. Since $\overline{\|\exists x\psi\|_{\oplus}^{\mathfrak{A}_k}} = \|\exists x\psi\|_{\oplus}^{\mathfrak{A}_j} \cap \bar{S}(\mathfrak{A}_k)$ (where the notation of Definition 9.5 is now used for $\mathfrak{A}_k \prec \mathfrak{A}_j$), it follows that $s \in \|\exists x\psi\|_{\oplus}^{\mathfrak{A}_k}$. Conversely, if $s \in \|\exists x\psi\|_{\oplus}^{\mathfrak{A}_k}$ then there exist $a, b \in A_k$ such that $\bar{s}_x^a \notin \|\psi\|_{\odot}^{\mathfrak{A}_k}$ and $\bar{s}_x^b \notin \|\psi\|_{\odot}^{\mathfrak{A}_k}$. Then, there exist $a, b \in A$ such that $\bar{s}_x^a \notin \|\psi\|_{\odot}^{\mathfrak{A}}$ and $\bar{s}_x^b \notin \|\psi\|_{\odot}^{\mathfrak{A}}$, by I.H. and $A_k \subseteq A$. Hence $s \in \|\exists x\psi\|_{\oplus}^{\mathfrak{A}}$.

(2.2) Assume that $s \in \|\exists x\psi\|_{\odot}^{\mathfrak{A}_k} = \widehat{\nabla x}(\|\psi\|_{\odot}^{\mathfrak{A}_k})$. Suppose that $\bar{s} \notin \widehat{\nabla x}(\|\psi\|_{\odot}^{\mathfrak{A}})$. Then, there exists $a \in A$ such that $\bar{s}_x^a \notin \|\psi\|_{\odot}^{\mathfrak{A}}$. Hence, there exists $j \in \mathbb{N}$ and $a \in A_j$ such that $\bar{s}_x^a \notin \|\psi\|_{\odot}^{\mathfrak{A}_j}$. If $j \leq k$ then $\bar{s}_x^a \notin \|\psi\|_{\odot}^{\mathfrak{A}_k}$ for some $a \in A_k$, by I.H., a contradiction (since $s \in \widehat{\nabla x}(\|\psi\|_{\odot}^{\mathfrak{A}_k})$). Otherwise, if $k < j$ then $\mathfrak{A}_k \prec \mathfrak{A}_j$, and so, by I.H., $s_x^a \notin \|\psi\|_{\odot}^{\mathfrak{A}_j}$ for some $a \in A_j$. From this, $s \notin \|\exists x\psi\|_{\odot}^{\mathfrak{A}_j}$ and so $s \notin \|\exists x\psi\|_{\odot}^{\mathfrak{A}_k}$ (given that $\mathfrak{A}_k \prec \mathfrak{A}_j$), a contradiction. Therefore, $\bar{s} \in \widehat{\nabla x}(\|\psi\|_{\odot}^{\mathfrak{A}}) = \|\exists x\psi\|_{\odot}^{\mathfrak{A}}$. Conversely, suppose that $\bar{s} \in \|\exists x\psi\|_{\odot}^{\mathfrak{A}} = \widehat{\nabla x}(\|\psi\|_{\odot}^{\mathfrak{A}})$. Then $\bar{s}_x^a \in \|\psi\|_{\odot}^{\mathfrak{A}}$ for every $a \in A$ and so $\bar{s}_x^a \in \|\psi\|_{\odot}^{\mathfrak{A}}$ for every

$a \in A_k$. By I.H., $s_x^a \in \|\psi\|_{\odot}^{\mathfrak{A}_k}$ for every $a \in A_k$, hence $s \in \widehat{\forall x}(\|\psi\|_{\odot}^{\mathfrak{A}_k}) = \|\exists x\psi\|_{\odot}^{\mathfrak{A}_k}$.

(2.3) The proof for $\|\exists x\psi\|_{\odot}^{\mathfrak{A}}$ follows from (2.1) and (2.2).

(3) If φ is $\forall x\psi$, then:

(3.1) Suppose that $\bar{s} \in \|\forall x\varphi\|_{\oplus}^{\mathfrak{A}} = \widehat{\exists x}(\|\psi\|_{\oplus}^{\mathfrak{A}}) - \widehat{\exists x}(\|\psi\|_{\odot}^{\mathfrak{A}})$. Then,

$$(*) \quad s \notin \widehat{\exists x}(\|\psi\|_{\odot}^{\mathfrak{A}_j}) \text{ for every } j \geq k.$$

Indeed, let $j \geq k$. Since $\bar{s} \notin \widehat{\exists x}(\|\psi\|_{\odot}^{\mathfrak{A}})$ then $\bar{s}_x^a \notin \|\psi\|_{\odot}^{\mathfrak{A}}$, for every $a \in A$. In particular, $\bar{s}_x^a \notin \|\psi\|_{\odot}^{\mathfrak{A}}$ for every $a \in A_j$. By I.H. for j , $s_x^a \notin \|\psi\|_{\odot}^{\mathfrak{A}_j}$, for every $a \in A_j$. Then $s \notin \widehat{\exists x}(\|\psi\|_{\odot}^{\mathfrak{A}_j})$, proving $(*)$. Now, since $\bar{s} \in \widehat{\exists x}(\|\psi\|_{\oplus}^{\mathfrak{A}})$ there exists $a \in A$ such that $\bar{s}_x^a \in \|\psi\|_{\oplus}^{\mathfrak{A}}$. Then, there exists $j \in \mathbb{N}$ and $a \in A_j$ such that $\bar{s}_x^a \in \|\psi\|_{\oplus}^{\mathfrak{A}}$. If $j \leq k$ then $s_x^a \in \|\psi\|_{\oplus}^{\mathfrak{A}_k}$, by I.H., and so $s \in \widehat{\exists x}(\|\psi\|_{\oplus}^{\mathfrak{A}_k})$. From this and $(*)$ it follows that $s \in \|\forall x\psi\|_{\oplus}^{\mathfrak{A}_k}$. Otherwise, if $k < j$, $s_x^a \in \|\psi\|_{\oplus}^{\mathfrak{A}_j}$ by I.H. for j . Then $s \in \widehat{\exists x}(\|\psi\|_{\oplus}^{\mathfrak{A}_j})$. From this and $(*)$, $s \in \|\forall x\psi\|_{\oplus}^{\mathfrak{A}_j}$. Given that $\mathfrak{A}_k \prec \mathfrak{A}_j$, it follows that $s \in \|\forall x\psi\|_{\oplus}^{\mathfrak{A}_k}$.

(3.2) The case for $\|\forall x\psi\|_{\odot}^{\mathfrak{A}_k}$ is treated analogously.

(3.3) The proof for $\|\forall x\psi\|_{\odot}^{\mathfrak{A}}$ follows easily from (3.1) and (3.2). $\square$

Let $\mathfrak{A}$ be a **QCiore**-structure for Θ . If $\emptyset \neq X \subseteq A$, we expand Θ to a new signature Θ_X by adding a new constant symbol c_a for each $a \in X$. In this way, we can expand $\mathfrak{A}$ to a **QCiore**-structure $\mathfrak{A}_X = (\mathfrak{A}, a)_{a \in X}$ for Θ_X where $c_a^{\mathfrak{A}_X} := a$. Given the map $h : A \rightarrow B$ where B is the universe of the **QCiore**-structure $\mathfrak{B}$ for Θ , then $\mathfrak{B}_{h(X)} = (\mathfrak{B}, h(a))_{a \in X}$ is the expansion of $\mathfrak{B}$ to a **QCiore**-structure for Θ_X where $c_a^{\mathfrak{B}_{h(X)}} = h(a)$.

More generally, we can choose a sequence $\vec{c} = c_1, \dots, c_n$ of new constant symbols that are not in Θ and consider the new signature $\Theta_{\vec{c}}$ which is the expansion of Θ obtained by adding the constant symbols $\vec{c}$. If $\mathfrak{A}$ is a **QCiore**-structure over Θ , we denote by $(\mathfrak{A}, \vec{a})$ the expansion of $\mathfrak{A}$ in the signature $\Theta_{\vec{c}}$ in such a way that $c_i^{(\mathfrak{A}, \vec{a})} = a_i$. If $\vec{x} = x_1, \dots, x_n$ is a sequence of n distinct variables then, for every $s \in S(\mathfrak{A})$ such that $s(x_i) = a_i$, we have that

- $t^{\mathfrak{A}}[s] = t(\vec{c})^{(\mathfrak{A}, \vec{a})}$ for all term $t(\vec{x})$
- $\|\varphi\|_{\mathfrak{A}}^{\mathfrak{A}}(s) = \|\varphi(\vec{c})\|_{\mathfrak{A}}^{(\mathfrak{A}, \vec{a})}$ for every formula $\varphi(\vec{x})$ in the language $\mathcal{L}(\Theta)$.

It is not difficult to prove the next useful lemma.

LEMMA 9.10 (Lemma on Constants). *Let $T \subseteq \text{Sent}(\Theta)$ be a theory and let $\varphi(\vec{x})$ be a formula in $\mathcal{L}(\Theta)$. Let $\vec{c}$ be a sequence of different constant symbols not appearing in T . Then, $T \Vdash_{\text{QCiore}} \varphi(\vec{c})$ iff $T \Vdash_{\text{QCiore}} \forall \vec{x}\varphi$.*

Let $\mathfrak{A}$ and $\mathfrak{B}$ be two **QCiore**-structures. A function $h : A \rightarrow B$ is an *elementary embedding* of $\mathfrak{A}$ into $\mathfrak{B}$, denoted by $h : \mathfrak{A} \prec \mathfrak{B}$, if, for all formulas φ and assignments $s \in S(\mathfrak{A})$, we have:

$$s \in \|\varphi\|_{\#}^{\mathfrak{A}} \text{ iff } h \circ s \in \|\varphi\|_{\#}^{\mathfrak{B}}$$

for each $\# \in \{\oplus, \odot, \ominus\}$. If $h : \mathfrak{A} \prec \mathfrak{B}$ and h is onto, we say that h is an *isomorphism* between $\mathfrak{A}$ and $\mathfrak{B}$. In such a case, we shall say that $\mathfrak{A}$ and $\mathfrak{B}$ are *isomorphic* and we shall write $\mathfrak{A} \simeq \mathfrak{B}$. It is easy to prove that $\mathfrak{A} \equiv \mathfrak{B}$ whenever $\mathfrak{A} \simeq \mathfrak{B}$.

The *elementary diagram* of $\mathfrak{A}$ is the theory $\text{Th}(\mathfrak{A}_A) := \{\varphi \in \text{Sent}(\Theta_A) : \mathfrak{A}_A \models_{\mathbf{QCiore}} \varphi\}$. As in classical model theory, the notion of elementary diagram is a tool used to characterize elementary embeddings.

PROPOSITION 9.11. *Let $\mathfrak{A}$ and $\mathfrak{B}$ be two **QCiore**-structures for Θ and let $f : A \rightarrow B$ be a function. The following conditions are equivalent:*

- (a) *f is an elementary embedding of $\mathfrak{A}$ into $\mathfrak{B}$,*
- (b) *there is an elementary extension $\mathfrak{C} \succ \mathfrak{A}$ and an isomorphism $g : \mathfrak{C} \rightarrow \mathfrak{B}$ such that $f \subseteq g$,*
- (c) *$(\mathfrak{B}, f(a))_{a \in A}$ is a **QCiore**-model of the elementary diagram of $\mathfrak{A}$.*

REMARK 9.12. *Let Θ , Θ_1 and Θ_2 be signatures such that $\Theta \subset \Theta_1$ and $\Theta \subset \Theta_2$, if $\mathfrak{A}$ and $\mathfrak{B}$ are **QCiore**-structures over Θ_1 and Θ_2 , respectively, we shall write*

$$\mathfrak{A} \equiv_{\Theta} \mathfrak{B}$$

to indicate that the Θ -reducts of $\mathfrak{A}$ and $\mathfrak{B}$ are elementary equivalent and, we shall write

$$f : \mathfrak{A} \prec_{\Theta} \mathfrak{B}$$

to indicate that f is an elementary embedding of $\mathfrak{A}|_{\Theta}$ into $\mathfrak{B}|_{\Theta}$.

THEOREM 9.13. *Let $\mathfrak{A}$ and $\mathfrak{B}$ be two **QCiore**-structures over Θ and Θ' , respectively, such that $\Theta \subset \Theta'$. If $\mathfrak{A} \equiv_{\Theta} \mathfrak{B}$, then there exists a **QCiore**-structure $\mathfrak{C}$ over Θ' such that $f : \mathfrak{A} \prec_{\Theta} \mathfrak{C}$ and $g : \mathfrak{B} \prec_{\Theta'} \mathfrak{C}$.*

Proof. Let $\Theta' := \Theta_{\mathfrak{A}} \cup \Theta_{\mathfrak{B}}$ (and suppose that $A \cap B = \emptyset$). Let us consider the theory

$$T := \text{Th}(\mathfrak{A}_A) \cup \text{Th}(\mathfrak{B}_B) \subseteq \text{Sent}(\Theta').$$

Suppose that there exists $T_0 \subseteq T$ (finite) which does not have models. By taking conjunctions, we have that there exist $\varphi(\vec{a}) \in \text{Th}(\mathfrak{A}_A) \cap T_0$ and $\psi(\vec{b}) \in \text{Th}(\mathfrak{B}_B) \cap T_0$ such that $\varphi \wedge \psi$ does not have **QCiore**-models. Then, no **QCiore**-structure over $\Theta'_{\mathfrak{B}}$ which satisfies $\psi(\vec{b})$ can be extended to a **QCiore**-structure with interpretations for the symbols $\vec{a}$ such that $\varphi(\vec{a})$ is **QCiore**-satisfied. Thus,

$$\begin{aligned} & \text{Th}(\mathfrak{B}_B) \models_{\mathbf{QCiore}} \neg\varphi(\vec{a}) \wedge \circ\varphi(\vec{a}) \\ \Rightarrow & \text{Th}(\mathfrak{B}_B) \models_{\mathbf{QCiore}} \forall \vec{x} (\neg\varphi(\vec{x}) \wedge \circ\varphi(\vec{x})) && \text{(Lemma on Constants)} \\ \Rightarrow & \mathfrak{B}_B \models_{\mathbf{QCiore}} \forall \vec{x} (\neg\varphi(\vec{x}) \wedge \circ\varphi(\vec{x})) \\ \Rightarrow & \mathfrak{A}_A \models_{\mathbf{QCiore}} \forall \vec{x} (\neg\varphi(\vec{x}) \wedge \circ\varphi(\vec{x})) \\ \Rightarrow & \mathfrak{A}_A \models_{\mathbf{QCiore}} \neg\varphi(\vec{a}) \wedge \circ\varphi(\vec{a}). \end{aligned}$$

Therefore $\mathfrak{A}_A \models_{\mathbf{QCiore}} \varphi(\vec{a}) \wedge \neg\varphi(\vec{a}) \wedge \circ\varphi(\vec{a})$. By Theorem 7.24, we have that T has a model $\mathfrak{C}$. Let us define the functions $f : A \rightarrow C$ and $g : B \rightarrow C$ such that $f(x) = x^{\mathfrak{C}}$ (for $x \in A$) and $g(y) = y^{\mathfrak{C}}$ (for $y \in B$). By Proposition 9.11, f and g are elementary embeddings. $\square$

THEOREM 9.14 (Robinson's Joint Consistency Theorem). *Let Θ_1 and Θ_2 be two signatures such that $\Theta = \Theta_1 \cap \Theta_2$. Suppose that*

- (i) $T \subseteq \text{Sent}(\Theta)$ is a maximal **QCiore**-consistent theory,
- (ii) $T_1 \subseteq \text{Sent}(\Theta_1)$ and $T_2 \subseteq \text{Sent}(\Theta_2)$ contain T , and
- (iii) both T_1 and T_2 have a **QCiore**-model.

Then, $T_1 \cup T_2$ has a **QCiore**-model.

Proof. Let $\mathfrak{A}_0$ be a **QCiore**-model of T_1 and $\mathfrak{B}_0$ be a **QCiore**-model of T_2 . Then, $\mathfrak{A}_0$ and $\mathfrak{B}_0$ are models of T and, since T is maximal **QCiore**-consistent, we have that $\mathfrak{A}_0 \equiv_{\Theta} \mathfrak{B}_0$. By Theorem 9.13 and Proposition 9.11, there exists a **QCiore**-structure $\mathfrak{B}_1$ such that $f_0 : \mathfrak{A}_0 \prec_{\Theta} \mathfrak{B}_1$. Since $(\mathfrak{A}_0, A_0) \equiv_{\Theta_{A_0}} (\mathfrak{B}_1, f_0(A_0))$, again by Theorem 9.13 and Proposition 9.11, there exists a **QCiore**-structure $\mathfrak{A}_1$ such that $g_0 : \mathfrak{B}_1 \prec_{\Theta} \mathfrak{A}_1$. Repeating this process, we can build a tower like the following:

$$\begin{array}{ccccccc}
 \mathfrak{A}_0 & \xrightarrow{\preceq} & \mathfrak{A}_1 & \xrightarrow{\preceq} & \mathfrak{A}_2 & \longrightarrow & \dots \\
 & \searrow f_0 & \uparrow g_0 & \searrow f_1 & \uparrow g_1 & & \\
 \mathfrak{B}_0 & \xrightarrow{\preceq} & \mathfrak{B}_1 & \xrightarrow{\preceq} & \mathfrak{B}_2 & \longrightarrow & \dots
 \end{array}$$

where $f_k : \mathfrak{A}_k \prec_{\Theta} \mathfrak{B}_{k+1}$ and $g_k : \mathfrak{B}_{k+1} \prec_{\Theta} \mathfrak{A}_{k+1}$ are such that

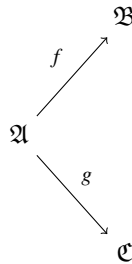
$$f_k \subseteq g_k^{-1} \subseteq f_{k+1}.$$

This is consequence of the characterization for embeddings given by means of elementary diagrams. By the Elementary Chain Theorem, $\mathfrak{A} = \bigcup \mathfrak{A}_k$ is a **QCiore**-model of T_1 and $\mathfrak{B} = \bigcup \mathfrak{B}_k$ is a **QCiore**-model of T_2 . Besides, $\bigcup f_k$ is an isomorphism of $\mathfrak{A}|\Theta$ onto $\mathfrak{B}|\Theta$. Then, $\mathfrak{B}$ is isomorphic to a **QCiore**-structure $\mathfrak{B}'$ such that $\mathfrak{A}|\Theta = \mathfrak{B}'|\Theta$. Let $\mathfrak{C}$ be the **QCiore**-structure on $\Theta_1 \cup \Theta_2$ such that

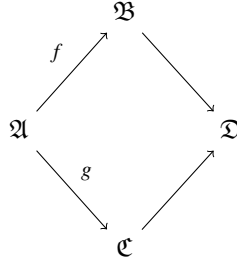
- $|\mathfrak{C}| = |\mathfrak{A}|$.
- If c is a constant symbol in Θ_2 , then $c^{\mathfrak{C}} := c^{\mathfrak{B}'}$.
- If c is a constant symbol in $\Theta_1 - \Theta_2$, then $c^{\mathfrak{C}} := c^{\mathfrak{A}}$.
- If f is a function symbol in Θ_2 , $f^{\mathfrak{C}} := f^{\mathfrak{B}'}$.
- If f is a function symbol in $\Theta_1 - \Theta_2$, $f^{\mathfrak{C}} := f^{\mathfrak{A}}$.
- If R is a predicate symbol in Θ_2 , $R^{\mathfrak{C}} := R^{\mathfrak{B}'} = \langle R_{\oplus}^{\mathfrak{B}'}, R_{\ominus}^{\mathfrak{B}'}, R_{\odot}^{\mathfrak{B}'} \rangle$.
- If R is a predicate symbol in $\Theta_1 - \Theta_2$, $R^{\mathfrak{C}} := R^{\mathfrak{A}} = \langle R_{\oplus}^{\mathfrak{A}}, R_{\ominus}^{\mathfrak{A}}, R_{\odot}^{\mathfrak{A}} \rangle$.

Then, $\mathfrak{C}$ is a model of $T_1 \cup T_2$. □

THEOREM 9.15 (Amalgamation). *Every diagram*



of elementary embeddings between QCiore-structures on Θ , can be completed to a commutative diagram



of elementary embeddings between **QCiore**-structures on Θ .

Proof. Let us restate the elementary diagram $Th(\mathfrak{B}, B)$ in the signature $\Theta_1 = \Theta(A \cup (B - f(A)))$. In this signature, $c_a^{\mathfrak{B}} = f(a)$, if $a \in A$. Analogously, we state the elementary diagram $Th(\mathfrak{C}, C)$ in the signature $\Theta_2 = \Theta(A \cup (C - g(A)))$ and, in this signature, $c_a^{\mathfrak{C}} = g(a)$ if $a \in A$. Then, $\Theta_1 \cap \Theta_2 = \Theta(A)$. In Θ_A we have a maximal **QCiore**-consistent theory $Th(\mathfrak{A}, A)$, and $Th(\mathfrak{B}, B)$ and $Th(\mathfrak{C}, C)$ are extensions of it in Θ_1 and Θ_2 , respectively. By Theorem 9.14, $Th(\mathfrak{B}, B) \cup Th(\mathfrak{C}, C)$ has a model $\mathfrak{D}$. Now, define the functions

$$f^* : \mathfrak{B} \rightarrow \mathfrak{D} \quad \text{and} \quad g^* : \mathfrak{C} \rightarrow \mathfrak{D}$$

$$x \mapsto c_x^{\mathfrak{D}} \quad \text{and} \quad x \mapsto c_x^{\mathfrak{D}}$$

Since $(\mathfrak{D}, f^*(B)) \models_{\mathbf{QCiore}} Th(\mathfrak{B}, B)$ and $(\mathfrak{D}, g^*(C)) \models_{\mathbf{QCiore}} Th(\mathfrak{C}, C)$ and by Proposition 9.11 we have that f^* and g^* are elementary embeddings.

Let $a \in \mathfrak{A}$, then $f^*(f(a)) = f^*(c_a^{\mathfrak{B}}) = c_a^{\mathfrak{D}} = g^*(c_a^{\mathfrak{C}}) = g^*(g(a))$. Therefore, $f^* \circ f = g^* \circ g$ (i.e., the diagram commutes). $\square$

THEOREM 9.16 (Craig's Interpolation Theorem). *Let φ, ψ be two sentences such that $\varphi \models_{\mathbf{QCiore}} \psi$. Then, there exists a sentence θ such that*

- (i) $\varphi \models_{\mathbf{QCiore}} \theta$ and $\theta \models_{\mathbf{QCiore}} \psi$.
- (ii) Every relation (excluding the identity), function or constant symbol that occurs in θ , also occurs in both φ and ψ .

In order to prove the Craig's Interpolation Theorem, we shall prove first the following technical results.

Given signatures Θ_1 and Θ_2 let $\Theta_0 := \Theta_1 \cap \Theta_2$, and let Θ'_0 be the extension of Θ_0 obtained by adding a countable set C of new constant symbols. Let $T \subseteq \text{Sent}(\Theta_1)$ and $U \subseteq \text{Sent}(\Theta_2)$. We say that a sentence $\theta \in \Theta_0$ *separates* the theories T and U if $T \models_{\mathbf{QCiore}} \theta$ and $U \models_{\mathbf{QCiore}} \neg\theta \wedge \circ\theta$. We say that T and U are *inseparable* if no sentence $\theta \in \text{Sent}(\Theta'_0)$ separates them.

LEMMA 9.17. *If T and U are inseparable, then both are **QCiore**-consistent.*

Proof. It is routine. $\square$

LEMMA 9.18. *If $T \cup \{\varphi\}$ and U are inseparable then: either $T \cup \{\varphi, \circ\varphi\}$ and U are inseparable, or $T \cup \{\varphi, \neg\varphi\}$ and U are inseparable.*

Proof. Suppose that $T \cup \{\varphi, \circ\varphi\}$ and U are separable, and that $T \cup \{\varphi, \neg\varphi\}$ and U are separable. Then, there are θ_1 and θ_2 such that

$$T \cup \{\varphi, \circ\varphi\} \models_{\mathbf{QCiore}} \theta_1, \quad U \models_{\mathbf{QCiore}} \neg\theta_1 \wedge \circ\theta_1$$

$$T \cup \{\varphi, \neg\varphi\} \Vdash_{\mathbf{QCiore}} \theta_2, \quad U \Vdash_{\mathbf{QCiore}} \neg\theta_2 \wedge \circ\theta_2$$

Since $\Vdash_{\mathbf{QCiore}} \varphi \rightarrow (\varphi \wedge \circ\varphi) \vee (\varphi \wedge \neg\varphi)$ and $\Vdash_{\mathbf{QCiore}} (\sim\alpha \wedge \sim\beta) \rightarrow \sim(\alpha \vee \beta)$ it follows that

$$T \cup \{\varphi\} \Vdash_{\mathbf{QCiore}} \theta_1 \vee \theta_2, \quad U \Vdash_{\mathbf{QCiore}} \neg(\theta_1 \vee \theta_2) \wedge \circ(\theta_1 \vee \theta_2).$$

From this, $T \cup \{\varphi\}$ and U are separable. $\square$

LEMMA 9.19. *If $T \cup \{\neg\varphi\}$ and U are inseparable then: either $T \cup \{\neg\varphi, \circ\varphi\}$ and U are inseparable, or $T \cup \{\varphi, \neg\varphi\}$ and U are inseparable.*

Proof. Suppose that $T \cup \{\neg\varphi, \circ\varphi\}$ and U are separable, and that $T \cup \{\varphi, \neg\varphi\}$ and U are separable. Then, there are θ_1 and θ_2 such that

$$T \cup \{\neg\varphi, \circ\varphi\} \Vdash_{\mathbf{QCiore}} \theta_1, \quad U \Vdash_{\mathbf{QCiore}} \neg\theta_1 \wedge \circ\theta_1$$

$$T \cup \{\varphi, \neg\varphi\} \Vdash_{\mathbf{QCiore}} \theta_2, \quad U \Vdash_{\mathbf{QCiore}} \neg\theta_2 \wedge \circ\theta_2$$

Since $\Vdash_{\mathbf{QCiore}} \neg\varphi \rightarrow ((\neg\varphi \wedge \circ\varphi) \vee (\varphi \wedge \neg\varphi))$ it follows that

$$T \cup \{\neg\varphi\} \Vdash_{\mathbf{QCiore}} \theta_1 \vee \theta_2, \quad U \Vdash_{\mathbf{QCiore}} \neg(\theta_1 \vee \theta_2) \wedge \circ(\theta_1 \vee \theta_2).$$

From this, $T \cup \{\neg\varphi\}$ and U are separable. $\square$

Analogously, we prove the next

LEMMA 9.20. *If $T \cup \{\circ\varphi\}$ and U are inseparable then: either $T \cup \{\varphi, \circ\varphi\}$ and U are inseparable or $T \cup \{\neg\varphi, \circ\varphi\}$ and U are inseparable.*

Proof. (of the Craig's Interpolation Theorem) Suppose that there does not exist a Craig interpolant θ of φ and ψ . We shall prove that it cannot happen that $\varphi \Vdash_{\mathbf{QCiore}} \psi$. For this, we shall construct a $\mathbf{QCiore}$ -model of $\varphi \wedge (\neg\psi \wedge \circ\psi)$. Let Θ_1 the signature with all the symbols that occur in φ and Θ_2 the signature with all the symbols that occur in ψ . Now, consider the signature

$$\Theta_0 := \Theta_1 \cap \Theta_2, \quad \Theta := \Theta_1 \cup \Theta_2.$$

Let Θ' be an extension of Θ that is obtained by adding a countable set C of new constant symbols and let

$$\Theta'_0 = \Theta_0 \cup C, \quad \Theta'_1 = \Theta_1 \cup C, \quad \Theta'_2 = \Theta_2 \cup C.$$

(1) $\{\varphi\}$ and $\{\neg\psi, \circ\psi\}$ are inseparable. Indeed, suppose that there exists $\theta(\vec{c}) \in \text{Sent}(\Theta'_0)$, where $\vec{c}$ is a tuple of new constants C and such that $\varphi \Vdash_{\mathbf{QCiore}} \theta(\vec{c})$ and $\{\neg\psi, \circ\psi\} \Vdash_{\mathbf{QCiore}} \neg\theta(\vec{c}) \wedge \circ\theta(\vec{c})$. Then, $\theta(\vec{c}) \Vdash_{\mathbf{QCiore}} \psi$ and, by the Lemma on Constants, we have that $\varphi \Vdash_{\mathbf{QCiore}} \forall \vec{x} \theta(\vec{x})$ and $\forall \vec{x} \theta(\vec{x}) \Vdash_{\mathbf{QCiore}} \psi$. Therefore, φ and ψ have a Craig interpolant.

Let $\varphi_0, \varphi_1, \varphi_2, \dots$ be an enumeration of all the sentences in Θ_1 and let $\psi_0, \psi_1, \psi_2, \dots$ be an enumeration of all the sentences in Θ_2 . Then, we shall construct two increasing sequences of theories in the following way

$$\begin{aligned} \{\varphi\} &= T_0 \subseteq T_1 \subseteq T_2 \subseteq \dots, \\ \{\neg\psi, \circ\psi\} &= U_0 \subseteq U_1 \subseteq U_2 \subseteq \dots, \end{aligned}$$

in $\text{Sent}(\Theta'_1)$ and $\text{Sent}(\Theta'_2)$ respectively, such that:

- if $T_m \cup \{\varphi_m\}$ and U_m are inseparable, then $\varphi_m \in T_{m+1}$,
- if $U_m \cup \{\psi_m\}$ and T_{m+1} are inseparable, then $\psi_m \in U_{m+1}$.

Now, let $T_\omega = \bigcup T_m$ and $U_\omega = \bigcup U_m$.

(2) T_ω and U_ω are inseparable.

Indeed, if there is $\theta \in \text{Sent}(\Theta'_0)$ such that $T_\omega \Vdash_{\mathbf{QCiore}} \theta$ and $U_\omega \Vdash_{\mathbf{QCiore}} \neg\theta \wedge \circ\theta$. By Theorem 7.24, there exist finite sets $\Delta \subseteq T_\omega$ and $\Gamma \subseteq U_\omega$ such that $\Delta \Vdash_{\mathbf{QCiore}} \theta$ and $\Gamma \Vdash_{\mathbf{QCiore}} \neg\theta \wedge \circ\theta$. Then, there is $k \in \omega$ such that

$$T_k \Vdash_{\mathbf{QCiore}} \theta, \quad U_k \Vdash_{\mathbf{QCiore}} \neg\theta \wedge \circ\theta.$$

Then T_ω and U_ω are both **QCiore**-consistent, by Lema 9.17.

(3) T_ω is a maximal **QCiore-consistent theory in Θ'_1 , and U_ω is a maximal **QCiore**-consistent theory in Θ'_2 .** Suppose that $\varphi_m \notin T_\omega$, $\neg\varphi_m \notin T_\omega$ and $\circ\varphi_m \notin T_\omega$. Then, $T \cup \{\varphi_m, \neg\varphi_m\}$, $T \cup \{\varphi_m, \circ\varphi_m\}$ and $T \cup \{\neg\varphi_m, \circ\varphi_m\}$, are separable from U_m . Then, there are $\theta_1, \theta_2, \theta_3 \in \text{Sent}(\Theta'_0)$ such that

$$\begin{aligned} T_\omega \Vdash_{\mathbf{QCiore}} \varphi_m \wedge \neg\varphi_m \rightarrow \theta_1, & \quad U_\omega \Vdash_{\mathbf{QCiore}} \neg\theta_1 \wedge \circ\theta_1, \\ T_\omega \Vdash_{\mathbf{QCiore}} \varphi_m \wedge \circ\varphi_m \rightarrow \theta_2, & \quad U_\omega \Vdash_{\mathbf{QCiore}} \neg\theta_2 \wedge \circ\theta_2, \\ T_\omega \Vdash_{\mathbf{QCiore}} \neg\varphi_m \wedge \circ\varphi_m \rightarrow \theta_3, & \quad U_\omega \Vdash_{\mathbf{QCiore}} \neg\theta_3 \wedge \circ\theta_3. \end{aligned}$$

Then, since $\Vdash_{\mathbf{QCiore}} (\varphi_m \wedge \neg\varphi_m) \vee (\varphi_m \wedge \circ\varphi_m) \vee (\neg\varphi_m \wedge \circ\varphi_m)$ and $\Vdash_{\mathbf{QCiore}} (\sim\theta_1 \wedge \sim\theta_2 \wedge \sim\theta_3) \rightarrow \sim(\theta_1 \vee \theta_2 \vee \theta_3)$ it follows that

$$T_\omega \Vdash_{\mathbf{QCiore}} \theta_1 \vee \theta_2 \vee \theta_3 \quad \text{and} \quad U_\omega \Vdash_{\mathbf{QCiore}} \neg(\theta_1 \vee \theta_2 \vee \theta_3) \wedge \circ(\theta_1 \vee \theta_2 \vee \theta_3),$$

and this contradicts the inseparability of T_ω and U_ω . Then, T_ω is a maximal **QCiore**-consistent theory in Θ'_1 . If $\varphi_m \in T_\omega$, then $\varphi_m \in T_m$. Now, $\varphi_r = \neg\varphi_m$ and $\varphi_s = \circ\varphi_s$. By Lemma 9.18 and the **Ciore**-consistency of T_ω , we can assert that $\varphi_r \in T_\omega$ or $\varphi_s \in T_\omega$.

If $\varphi_r = \neg\varphi_m \in T_\omega$, then $\varphi_r \in T_r$. If $\varphi_s = \circ\varphi_s$, by Lemma 9.19 and the **Ciore**-consistency of T_ω , we have that $\varphi_m \in T_\omega$ or $\varphi_s \in T_\omega$.

If $\varphi_s = \circ\varphi_m \in T_\omega$, then $\varphi_s \in T_s$. If $\varphi_r = \neg\varphi_s$, by Lemma 9.20 and the **Ciore**-consistency of T_ω , we have that $\varphi_m \in T_\omega$ or $\varphi_r \in T_\omega$.

In a similar way, we prove the maximality of U_ω .

(4) $T_\omega \cap U_\omega$ is a maximal **QCiore-consistent theory in Θ'_0 .** Let σ be a sentence of Θ'_0 . By (3) and the inseparability, we have that either

$$\{\sigma, \neg\sigma\} \subseteq T_\omega \cap U_\omega \quad \text{or} \quad \{\sigma, \circ\sigma\} \subseteq T_\omega \cap U_\omega \quad \text{or} \quad \{\neg\sigma, \circ\sigma\} \subseteq T_\omega \cap U_\omega.$$

Then, by the Robinson's Joint Consistency Theorem, $T_\omega \cup U_\omega$ has a model. Therefore, $\{\varphi, \neg\varphi, \circ\varphi\}$ has a model and then it cannot be the case that $\varphi \Vdash_{\mathbf{QCiore}} \psi$. $\square$

§10. Conclusions. In the present article we have developed a model theory framework that was used successfully for studying a first-order version of the paraconsistent 3-valued logic **Ciore** called **QCiore**. As discussed in §7.1, **QCiore** is more faithful to the spirit of **Ciore** than the first-order version of **Ciore** called **LF12*** presented in Carnielli et al. (2000). The model theory for **QCiore** proposed here strongly relies on the notion of partial (or pragmatic) structures presented in §6. Based on it, it was shown in which manner classical

model theory can be adapted to this (more general) setting. In particular, important classical theorems of classical model theory were obtained in this framework, such as: *Tarski and Vaught Theorem* (generalizing the *Downward Löwenheim-Skolem-Tarski Theorem*), *Robinson's Joint Consistency Theorem*, *Amalgamation Theorem* and *Craig's Interpolation Theorem*. It is worth mentioning that the concept of pragmatic structure can be used to study other 3-valued **LFI**s. For instance, all the results obtained in D'Ottaviano (1982, 1985a,b, 1987) for the first-order version of **J3** can be recast in this framework.

An interesting task to be done is trying to adapt this framework to suitable first-order versions of some interesting members of the family of 8K 3-valued **LFI**s mentioned in the Introduction, to which **LFI1** and **Ciore** belong, or even to other 3-valued logics.

A first-order extension of **mbC** called **QmbC** was proposed in Carnielli et al. (2014), and it was additionally studied in (Carnielli & Coniglio, 2016, Chapter 7). In Ferguson (2018) it was obtained an important model-theoretic result on **QmbC**, the Keisler-Shelah Theorem, by using a novel technique of model-theoretic “atomization”. This theorem should be valid in the stronger first-order **LFI**s extending it like **LPT1** and **QCiore**. This being so, it would be interesting to adapt the proof given in Ferguson (2018) to the semantical framework of partial structures.

Acknowledgments. We would like to thank the anonymous referees for their valuable suggestions and comments, which helped to improve the final version of the article. Coniglio was financially supported by an individual research grant from CNPq, Brazil (308524/2014-4). Gomez-Pereira was financially supported by a doctoral fellowship from CONICET, Argentina. This work was partially supported by project 24/ZL15 at Universidad Nacional del Sur, Argentina.

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